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Alzheimer's disease target and drug discovery by leveraging multiomics and electronic health data

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Despite prolific genomic data from individuals with Alzheimer's disease (AD), translation into treatments has lagged. Here we applied Mendelian randomization to multiple AD genome-wide association studies and functional multiomics datasets to nominate druggable targets, identifying 19 targets for individuals of European ancestry and 7 for African ancestry. This included epoxide hydrolase 2 (*EPHX2*), for which we identified a protective missense variant (p.Arg287Gln). Experiments in induced pluripotent stem cell-derived neurons and cerebral organoids from patients with AD validated the protective role of p.Arg287Gln via reduced levels of *EPHX2* and phosphorylated tau (pTau-181). Pharmacological inhibition of *EPHX2* improved cognitive performance in the 5xFAD mouse model of AD. Finally, by leveraging electronic health data from 111,680 patients with mild cognitive impairment, we identified 12 drugs associated with reduced AD incidence, including trazodone (targeting ADRA1A) and baclofen (targeting GABBR1). Thus, integrating multiomics data, functional validation and electronic health data identifies new approaches for treating AD.

Alzheimer's disease (AD) and AD-related dementias are a major global health challenge, with approximately 150 million people predicted to be afflicted worldwide by 2050[1]. The failure rate for AD clinical trials is 99.6% (ref. 2), and only two disease-modifying drugs (lecanemab[3] and donanemab[4]) have been approved by the US Food and Drug Administration (FDA) in the past two decades.

AD is highly heritable (90% for early-onset AD and 58–79% for late-onset AD)[5], and genetically supported targets double the success rate for drug development[6]. Thus, therapeutically modulating genetic risk genes holds promise for developing disease-modifying treatments[7]. Indeed, high-throughput DNA and RNA sequencing technologies have generated large-scale genetic and genomic data in the Alzheimer's Disease Sequencing Project[8] and the Alzheimer's Disease Neuroimaging Initiative[9]. However, genetics and genomics data have not yet been fully utilized to guide the development of disease-modifying therapeutics for AD. For example, the systematic identification and characterization of genetic risk factors from ethnically diverse populations could bolster this effort[10]. Utilization of

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comprehensive real-world electronic health data holds similar potential by enabling efficient hypothesis testing through massive amounts of patient data[11]. In particular, real-world patient data offer large patient populations for detecting small differences, along with recorded clinical factors that reduce bias from patient recall. These features enable high-dimensional covariate adjustment that minimizes confounding factors and enables rapid screening for effective therapies in AD and AD-related dementia[12].

With this in mind, we posited that integration of extensive genetic data with real-world patient data would further boost the likelihood of identifying disease-modifying drug targets and treatments for AD[13]. We therefore performed Mendelian randomization (MR) analysis to screen 1,229 druggable targets with known US FDA-approved drugs or molecules under clinical investigation for AD. In total, we analyzed genetic data from AD cases and controls across seven AD genome-wide association study (GWAS) datasets for both European ancestry (EA) and African ancestry (AA). We identified 19 potential AD drug targets in EA and 7 in AA. We also performed drugome-wide association studies (DWASs) to validate our MR-positive targets from real-world patient databases and identified 12 drugs associated with reduced AD incidence in individuals with mild cognitive impairment (MCI). In summary, we present a drug target compendium supported by genetic evidence and real-world patient data for AD in individuals of EA or AA and offer a platform approach for using existing genetic and large real-world patient data to identify potential treatments for AD.

Results

An AD drug target compendium

To identify drug targets and medicines that could be repurposed for AD, we developed a genome–phenome association analytic framework composed of three key components: (1) genetics-based MR analysis (Fig. 1a); (2) experimental validation in neurons and brain organoids derived from patient-derived induced pluripotent stem (iPS) cells, as well as in a transgenic mouse model of AD (5xFAD) (Fig. 1b); and (3) real-world patient data-based drug repurposing validation (Fig. 1c).

We conducted MR analyses to identify potential drug targets from 1,229 druggable proteins (Extended Data Fig. 1). A protein was considered druggable if targeted by known FDA-approved medicines or investigational molecules with binding affinity lower than 1 μM (Methods). Next, we selected single nucleotide polymorphisms (SNPs) as valid instrumental variables (IVs; false discovery rate (FDR)-adjusted P values < 0.05 and F-statistics > 10) for each protein-coding gene from three protein quantitative trait loci (pQTL) and nine gene expression quantitative trait loci (eQTL) datasets (Fig. 1, Extended Data Fig. 1 and Supplementary Table 1) derived from the Religious Orders Study/Memory and Aging Project (ROSMAP)[14], MetaBrain[15] and Mayo Clinic Biobank[16]. We also gathered seven AD GWAS summary statistics datasets from EA or AA individuals (Supplementary Table 2) and implemented five complementary MR models[17,18] (Extended Data Fig. 1; Methods) using multiple lines of evidence to strengthen our findings.

The DWAS analysis provides real-world patient data for drugs with targets prioritized by MR analyses. However, this approach is limited to FDA-approved drugs with sufficient participants in the patient database, leaving many MR-prioritized targets unvalidated. Therefore, we performed experimental validation of top MR-prioritized targets using iPS cell-derived neurons and brain organoids from patients with AD (Fig. 1), complemented by in vivo validation in 5xFAD mice for compounds at the preclinical or clinical investigation stage.

Alzheimer's MR score prioritized druggable targets

Considering the multiple genetic and genomic datasets and methods used in MR analysis, we used a consensus Alzheimer's MR score (alzMR score) to quantify the sensitivity of genetic evidence from MR (Supplementary Methods). We performed simulations to illustrate that the alzMR score (Supplementary Fig. 1) accurately identifies genes with the strongest genetic evidence and distinguishes between pairs of genes with similarly strong evidence. The Cauchy combination test (Supplementary Methods) further validated the robustness of the alzMR score across seven GWAS datasets from EA, despite partial participant overlapping among them (Supplementary Table 3). Notably, the alzMR score demonstrated sensitivity in distinguishing genetic signals when comparing clinically diagnosed AD GWAS datasets with proxy AD datasets (r = 0.69; Extended Data Fig. 2). This suggests that the alzMR score captures consistent signals across subsequent AD GWAS studies while accounting for variations in study population characteristics and diagnosis criteria.

We first applied the alzMR score to prioritize genes with likely causal effects on AD based on MR results significant at FDR < 0.05. Genes with alzMR score > 0 were classified as antagonistic targets, for which inhibition could provide therapeutic benefit, whereas genes with alzMR score < 0 were classified as agonistic targets, for which activation could provide therapeutic benefit. We prioritized drug targets that met the following conditions: (1) AD clinical trials completed or ongoing for the target; (2) validation of the FDA-approved drugs for the target using DWAS analysis; (3) target colocalization with posterior probabilities of hypothesis 4 (PP.H4) > 0.5; and (4) targets having fine-mapping of causal gene sets (FOCUS[19]) with posterior inclusion probabilities (PIPs) > 0.5. These criteria identified 73 drug targets associated with elevated (39 antagonistic targets) or reduced (34 agonistic targets) risk of AD with EA at FDR < 0.05 (Fig. 2 and Supplementary Table 4), based on Wald ratio and inverse-variance-weighted (IVW) methods. Angiotensin-converting enzyme (ACE; alzMR score = −199.12) and eukaryotic translation initiation factor 4E (alzMR score = −111.51) were the top 2 agonistic targets (Fig. 2 and Supplementary Table 4). ADAM metallopeptidase domain 10 (ADAM10; alzMR score = 133.56), lactamase-β (alzMR score = 165.47), ERCC excision repair 2 (alzMR score = 126.20), epoxide hydrolase 2 (EPHX2; alzMR score = 195.69) and epidermal growth factor receptor (EGFR; alzMR score = 121.84) were the top 5 AD antagonistic targets (FDR < 0.05; Fig. 2 and Supplementary Table 4). The MR results of agonistic and antagonistic drug targets were consistent across different MR methods (Extended Data Fig. 3). The top druggable targets with a stronger alzMR score were also replicated in an independent AD GWAS[20] with clinically diagnosed AD cases, including EPHX2, ACE, ADAM10, PRKCB, FOLH1, EGFR, CHRNE, TAS2R60 and CD38. Among them, ACE (PP.H4 = 0.77), ADAM10 (PP.H4 = 0.71) and CTSH (PP.H4 = 0.97) showed colocalization between pQTL in the cortex and AD GWAS loci[21] (Supplementary Fig. 2 and Supplementary Table 5). The eQTL for ADAM10 (PP.H4 = 0.98), CTSH (PP.H4 = 0.84) and EGFR (PP.H4 = 0.99) from multiple brain regions also colocalized with genome-wide significant AD GWAS loci. We also fine-mapped eight targets in AD (FOCUS PIP > 0.5; Supplementary Fig. 3 and Supplementary Table 6), including ACE, MAPT, EGFR, LACTB, EPHX2, ADAM10, PTGDR and DMPK.

Next, we examined top-prioritized MR targets using 17 cell-type-specific eQTL datasets and identified 20 targets (for example, APH1B, EGFR, ADAM10, PTK2B, EPHX2, KCNN4, CTSH and ACE) with significant MR results (FDR < 0.05) in at least three AD GWAS and cell-type-specific eQTL pairs (Supplementary Fig. 4). The most significant targets were identified in excitatory neurons, and colocalization of APH1B (PP.H4 = 0.77) and CTSH (PP.H4 = 0.99) was validated in four AD GWAS datasets (Supplementary Fig. 2). We also identified two druggable targets (KCNN4 and PTK2B) with significant MR scores in microglia and bulk human brain eQTL and pQTL datasets (Figs. 2 and 3). Colocalization between AD GWAS and microglia chromatin accessibility QTL has been independently confirmed for the *KCNN4* locus in a microglia-specific regulome analysis[22].

We also conducted MR analysis using an AD GWAS dataset of AA ancestry (2,784 AA AD cases and 5,222 AA control individuals[23] (Fig. 1). We first ranked putative targets for AA using alzMR score and defined targets in AA using MR P < 0.05 in at least three xQTL × AA AD

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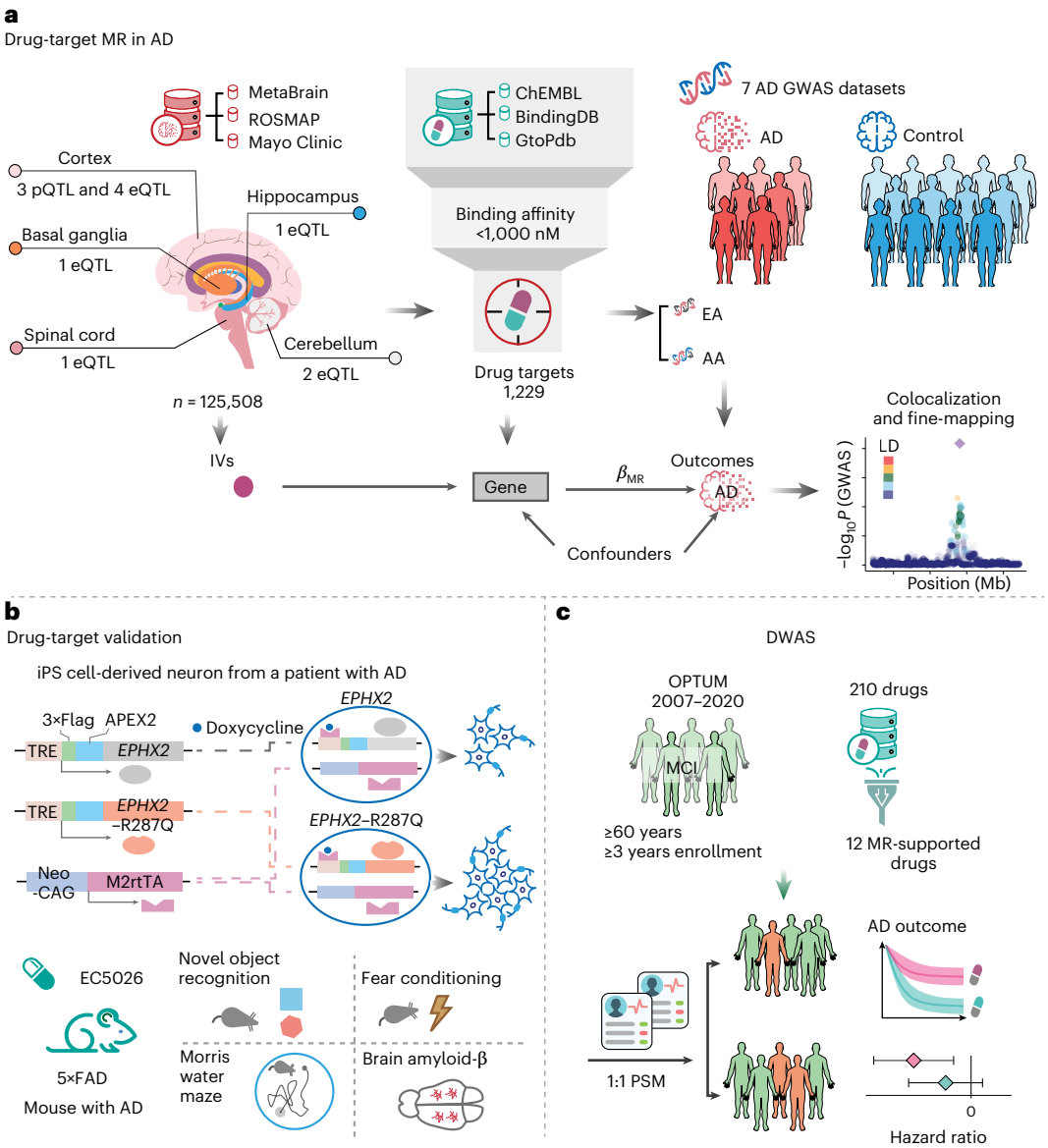


Fig. 1 | Experimental pipeline overview. a, A framework illustrating MR analysis for AD. The MR analysis workflow includes two parallel parts. Top: steps of MR-based drug target identification and validation. IVs corresponding to 1,229 druggable targets were selected from 3 pQTL and 9 eQTL datasets across 5 brain regions. In total, 7 GWAS summary statistics datasets, comprising 275,540 AD cases and 1.55 million controls of African American and European American ancestry, were used as outcome datasets (Methods). Five MR methods were used to enhance the reproducibility and rigor of drug target findings. **b**, Neurons from healthy and patient-derived iPS cells and a transgenic mouse model of AD (5xFAD) were used to validate drug targets. **c**, A framework illustrating DWASs. In total, 210 highly prescribed drugs were evaluated after adjusting for various confounding factors (such as age, sex, race and disease comorbidities) using 1:1 propensity score matching (PSM). LD, linkage disequilibrium.

GWAS pairs. This prioritized five antagonistic (dihydrofolate reductase (DHFR), SLC7A11, TRPV3, ABL1 and SETD7) and two agonistic drug targets (ALDH2 and PPARG) for AD in AA individuals ($P < 0.05$; Fig. 2 and Supplementary Table 7) by eQTL IVs.

AD target discovery from GWASs with EA

We next focused on 19 high-confidence AD drug targets (top 2% of 1,229 druggable targets) in EA (Fig. 3) based on three factors: (1) MR evidence (Fig. 2); (2) consistent direction of alzMR scores with the experimentally reported mechanism of action (MoA) of corresponding drugs[24]; and (3) targets with druggable pockets based on protein three-dimensional structural analysis (Supplementary Fig. 5). We found four antagonistic drug targets (0.3% of 1,229 targets) with the strongest alzMR scores (FDR < 0.05; Supplementary Fig. 6) across six AD GWAS datasets in EA, including EPHX2, EGFR, PRKCB and FOLH1 (Fig. 3a).

We classified the 19 alzMR-supported targets into seven therapeutic categories based on experimentally reported MoAs[2,25] (Fig. 3b and Supplementary Table 8). Anti-amyloid and neuromodulation were the top 2 MoA categories, accounting for 47% ($n = 9$) and 42% ($n = 8$) of the 19 targets, respectively (Supplementary Fig. 7a). Most targets in neuromodulation ($n = 5$; DRD2, CHRNE, ADRA1A, PRKCB and GABBR1) and amyloid ($n = 3$; BACE2, APH1B and CACNA1D) are annotated as a single MoA (Fig. 3b and Supplementary Fig. 7a). The remaining AD drug targets modulated inflammation (37%; $n = 7$; EPHX2, NR3C1, FPR1, FOLH1, CD38, plasminogen (PLG) and PARP10), metabolism (26%; $n = 5$; EPHX2, FOLH1, CD38, PPAR10 and VKORC1) and vasculature (16%; $n = 3$; VKORC1, PLG and EPHX2) pathways, with most spanning multiple MoAs. Both antagonistic targets (EPHX2 and FOLH1) contribute to more than four AD MoAs (anti-inflammation[26] and metabolic pathways[26]; Fig. 3b), beyond amyloid-β neurotoxicity[27,28] and tauopathy[28]. Thus,

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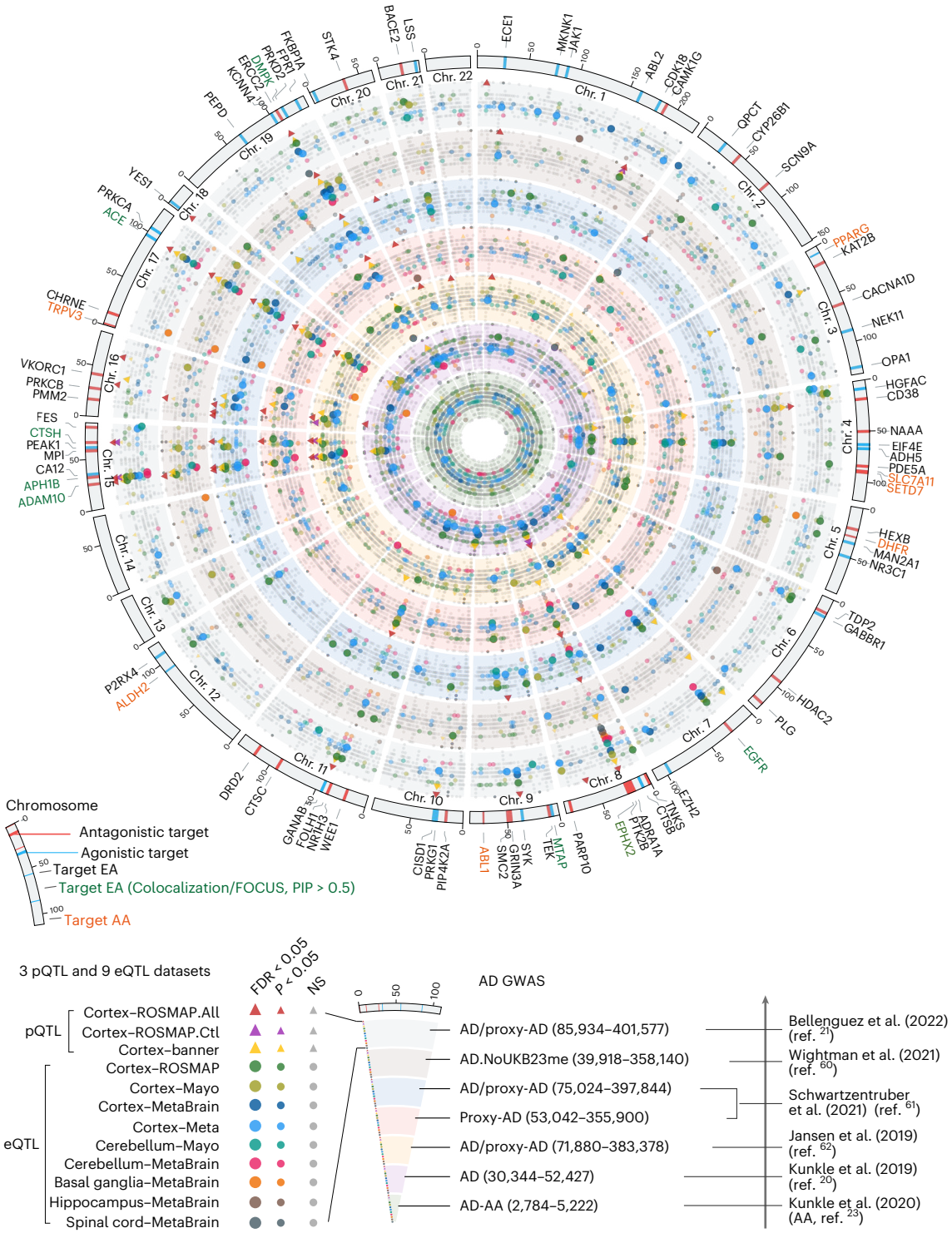


Fig. 2 | A drug target compendium for AD supported by human genetic evidence in both EA and AA. Circle plot illustrates MR (IVW method) results for 1,229 druggable targets across three pQTL datasets (triangles), nine eQTL datasets (circles) and seven AD GWAS datasets (distinguished by different background colors; six EA[20,21,60–62] and one AA[23]). *P* values are two-sided, derived from a Wald test of the IVW MR estimate against the standard normal distribution; multiple testing was controlled by the Benjamini–Hochberg FDR. Larger colored shapes denote FDR-corrected significance < 0.05; smaller colored shapes denote $P < 0.05$; gray shapes denote nonsignificant MR results. The outer circle shows chromosome position, with tick labels denoting the rank of each drug target in human hg19 genome reference. Blue lines in the chromosome circle indicate agonistic targets, whereas red lines indicate antagonistic targets. The 73 putative targets for EA include at least one MR result with FDR < 0.05. Green text highlights druggable targets for EA that were positive for colocalization or FOCUS fine-mapping (PP.H4 or PIP > 0.5). The seven suggestive targets for AA are highlighted in orange text at $P < 0.05$ across at least three MR tests with the top alzMR score. NS, nonsignificant.

genetic data support potent AD drug targets that act on multiple biological pathways extending outside of amyloidopathy and tauopathy.

We next evaluated 17 FDA-approved drugs and 13 investigational molecules corresponding to the 19 AD targets identified by alzMR score in EA (Fig. 3c and Supplementary Table 9; Methods). The 17 approved drugs were classified into nine anatomical therapeutic chemical categories, the top 3 of which were nervous system (for example, drug–target pairs: galantamine–CHRNE), cardiovascular system (amlodipine–CACNA1D) and antineoplastic and immunomodulating agents (daratumumab–CD38). We found that one FDA-approved AD drug (galantamine, targeting CHRNE; alzMR score = 46.53) had significant alzMR score for AD. We also identified 14 investigational molecules corresponding to ten targets (nine antagonistic and one agonistic), such as EC5026 and AR-9281 for EPHX2-encoded soluble epoxide hydrolase (sEH) and 2-(phosphonomethyl)pentanedioic acid for FOLH1 (Fig. 3c).

Among 30 drugs/molecules, seven across six targets are currently or have been tested in AD clinical trials (Fig. 3c and Supplementary Table 9). Phase I AD clinical trials primarily tested anti-amyloid agents, including AZD-3839 (ClinicalTrials.gov identifier: NCT01348737) and LY-2811376 (NCT00838084), small-molecule BACE1 inhibitors that also engage the closely related paralog BACE2 owing to shared active-site homology and begacestat (NCT00959881) as a γ-secretase inhibitor acting on the complex subunit APH1B. These observations reveal that MR analysis reliably identifies existing targets under AD clinical trials (Fig. 3c). Our alzMR scores also identified targets harboring two drugs currently under investigation in phase III AD trials, including prednisone (targeting NR3C1; NCT00000178) and AR1001 (a selective inhibitor of phosphodiesterase 5 (PDE5); NCT05531526).

EPHX2 is a potent anti-inflammatory target supported by AD GWAS

EPHX2, encoding epoxide hydrolase, breaks down neuronal protective metabolites known as epoxy fatty acids[29] and induces neuroinflammation and synapse damage in AD (Figs. 3b–5). EPHX2 was one of the strongest antagonistic drug targets prioritized by the alzMR score. Elevated EPHX2 protein expression (pQTL IVs ($n = 7$) derived from ROSMAP) was significantly associated with elevated AD risk (β (IVW MR) = 0.190; MR $P = 7.55 \times 10^{-8}$; Fig. 4a). We replicated these results across five other AD GWAS datasets to reduce the potential effects of horizontal pleiotropy and outlier IVs (Fig. 4a and Supplementary Fig. 8).

We next conducted conditional analyses and further functional genomic validation using human-brain single-nucleus assay for transposase-accessible chromatin sequencing (ATAC–seq) and Hi-C followed by chromatin immunoprecipitation[30]. *EPHX2* pQTL SNP rs2741342 (conditional $P = 2.49 \times 10^{-8}$) was identified as independent with lead SNP rs1532278 ($P_{GWAS} = 5.72 \times 10^{-25}$) in the *CLU* locus (Fig. 4b, Extended Data Fig. 4a,c and Supplementary Table 10). Additionally, joint association analysis (Supplementary Methods) found that SNPs rs1532278 (*CLU*), rs751141 (*EPHX2*), rs2741342 (*EPHX2*) and rs73223431 (*PTK2B*) have independent genome-wide significant associations with AD (Extended Data Fig. 4b and Supplementary Table 11). These include the *EPHX2*–p.Arg287Gln coding variant rs751141 in Bellenguez et al.[21] GWAS dataset (joint estimated $P = 1.43 \times 10^{-8}$) and *EPHX2* rs2741342 in Kunkle.Starge2[20] dataset; joint estimated $P = 3.48 \times 10^{-8}$). Human hippocampal ATAC–seq analysis further showed that an enhancer harboring the GWAS-assigned AD-risk SNP rs1532278 in *CLU* was linked to active promoters of *EPHX2* (Supplementary Fig. 9). Consistent with this regulatory connection, CRISPR interference (CRISPRi) of rs1532278 reduced both *CLU* and *EPHX2* expression in iPS cell-derived neurons, with no effect on *PTK2B* expression (Fig. 4c and Extended Data Fig. 4d). However, CRISPRi of SNP rs2741342 only reduced *EPHX2* expression, with no effect on *CLU* and *PTK2B* (Fig. 4c).

We also found a protein-coding variant of rs751141 (p.Arg287Gln (R287Q)) associated with reduced EPHX2 protein level ($\beta = -0.291$; $P_{pQTL} = 5.50 \times 10^{-16}$) and protection from AD in EA[21] ($\beta = -0.096$; $P_{GWAS} = 1.08 \times 10^{-11}$; Fig. 4b and Supplementary Fig. 10). These findings, together with GWAS conditional analysis and protein structural simulations (Supplementary Fig. 11), further support EPHX2 as a potent antagonistic target for AD. We also inspected functional roles of *EPHX2*–p.Arg287Gln using patient iPS cell-derived neurons and cerebral organoids. Using the transcription activator-like effector nuclease (TALEN) system[31] to insert a doxycycline-inducible overexpressed *EPHX2* or *EPHX2*–p.Arg287Gln at the *AAVS1* locus in iPS cell-derived forebrain neurons from patients with AD (Supplementary Fig. 12), we identified reduced phosphorylated tau (pTau) in *EPHX2*–p.Arg287Gln iPS cell-derived neurons ($P = 0.003$; Fig. 4d,e) and cerebral organoids ($P = 0.044$; Fig. 4f,g) from patients with AD. Analysis of available cerebrospinal fluid (CSF) biomarker GWAS data[32] showed that protective *EPHX2* rs751141 variant is nominally associated with lower pTau levels in individuals with abnormal amyloid levels ($P = 0.034$; Supplementary Fig. 13), whereas no association was observed in those with normal amyloid levels ($P = 0.69$).

Next, we performed a prime editing experiment[33] to test loss-of-function effect in three replicates of iPS cell-derived neurons from patients with AD. Prime editing was used to edit *EPHX2* rs751141 from G (Arg [CGG]) to A (Gln [CAG]) in the iPS cell genome of a patient with AD. We selected single colonies for each mutagenesis line and validated with Sanger sequencing (Fig. 4h). The G to A mutation significantly reduced EPHX2 protein levels in iPS cell-derived neurons of a patient with AD (Fig. 4h). Compared to wild-type lines (*EPHX2* G/G), the *EPHX2* A/A mutation increased neuronal differentiation and survival pathways (Fig. 4i,j). Notably, the *EPHX2* A/A mutation activated the PI3K–Akt pathway, which inhibits GSK-3β, a major tau kinase, consistent with reduced pTau by *EPHX2*–p.Arg287Gln mutation (Fig. 4f). In summary, these functional observations indicate that *EPHX2* is a likely causal gene for AD, and that the protective mutation p.Arg287Gln is associated with reduced pTau levels and neuroprotection in AD.

Pharmacological EPHX2 inhibition reduces pro-inflammatory metabolites and improves cognition in 5xFAD mice

We next tested whether pharmacological inhibition of EPHX2 has a beneficial effect in 5xFAD mice (Fig. 5). Specifically, we evaluated the effects of EC5026 (a picomolar EPHX2 inhibitor in phase I trials; ClinicalTrials.gov IDs: NCT06438471, NCT04228302 and NCT04908995) (Fig. 5a). We observed that EC5026 treatment significantly reduced amyloid-β pathology (Fig. 5b,d) and microglia-associated inflammation (Fig. 5c,d) while enhancing lysosomal function (Extended Data Fig. 5a). Single-cell analysis further revealed that EC5026 blocked multiple pro-inflammatory pathways in microglia (Fig. 5h,i and Extended Data Fig. 5b) and enhanced neuronal synapse pathways in excitatory neurons and oligodendrocytes (Extended Data Fig. 5c), consistent with findings from patient iPS cell-derived neurons (Fig. 4i,j).

We also evaluated the effects of EPHX2 inhibition on learning and hippocampal-dependent short-term and long-term memory. The novel object recognition test showed that EC5026 treatment significantly improved long-term recognition memory compared to vehicle-treated 5xFAD mice ($P = 0.014$; Fig. 5e) at 9 months. EC5026-treated mice also showed improved contextual fear memory compared to vehicle-treated groups ($P = 0.033$; Fig. 5f). In the Morris water maze, mice treated with EC5026 showed significantly reduced escape latency on training day 5 compared to the vehicle-treated group ($P = 0.004$; Fig. 5g) and increased time in target quadrant ($P = 0.048$). Taken together, pharmacologic inhibition of EPHX2 by EC5026 protected 5xFAD mice.

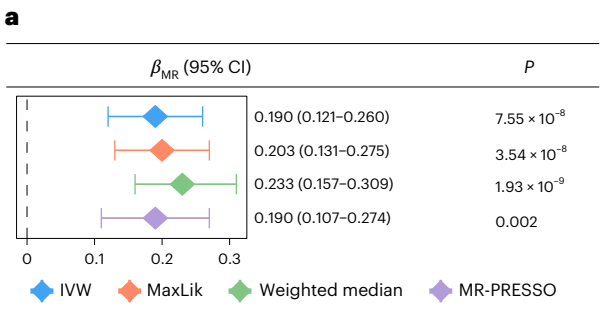
We next performed pharmacokinetic and brain target engagement analyses of EC5026 in 5xFAD mice. EC5026 achieved therapeutic concentrations in cortical regions 2 h after dosing with approximately 15% brain penetration (Extended Data Fig. 6a,b), and liquid chromatography–tandem mass spectrometry analysis confirmed on-target EPHX2 enzyme inhibition with reduced cortical and plasma levels of

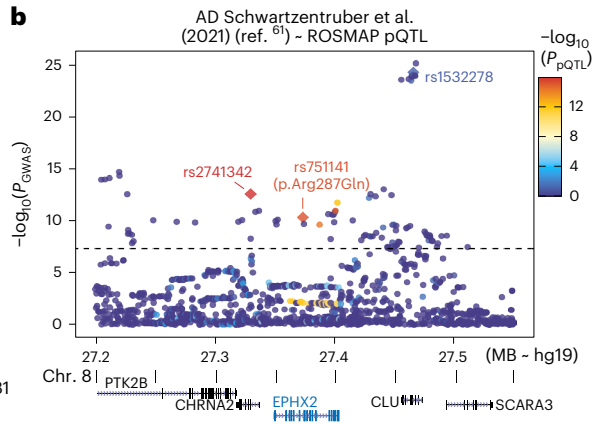
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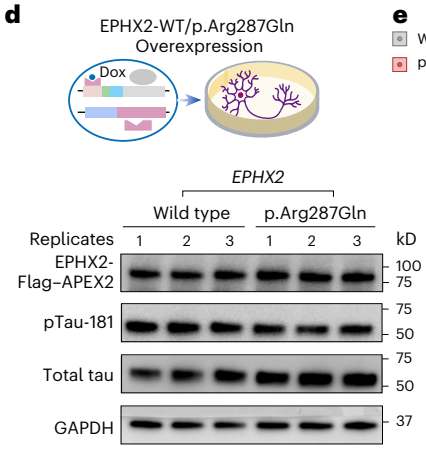


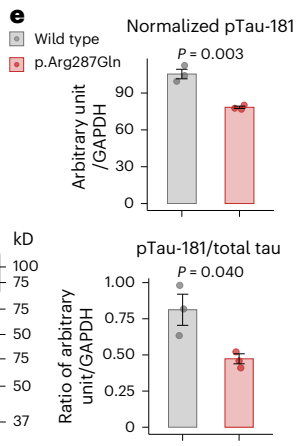
Fig. 3 | Support of human genetic evidence for AD drug target in EA population. a, Heat map shows the β_{MR} coefficient score of 19 drug targets for EA[20,21,60–62]. $\beta_{MR} > 0$ indicates an antagonistic target; $\beta_{MR} < 0$ indicates an agonistic target. Asterisks (**) denote significance at FDR < 0.05, corrected for multiple testing using the Benjamini–Hochberg method. NA, not available for MR analysis due to the lack of valid instrumental variables. **b**, Distribution of AD drug targets across MoA categories. The left panel shows the distribution of the 19 targets across seven MoA categories: amyloid, tau, inflammation, neuromodulation, microglia, metabolism and vasculature. The right bar graph presents the alzMR score of 19 drug targets: ruby denotes antagonistic targets and green denotes agonistic targets. **c**, Sankey plot shows the relationship between repurposed drugs and their alzMR-score-predicted targets for AD. Dark green indicates that the drug acts as an agonist of the target; ruby indicates that the drug acts as an inhibitor/antagonist of the target.

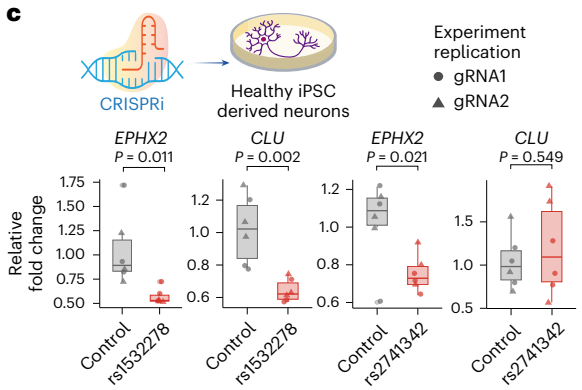
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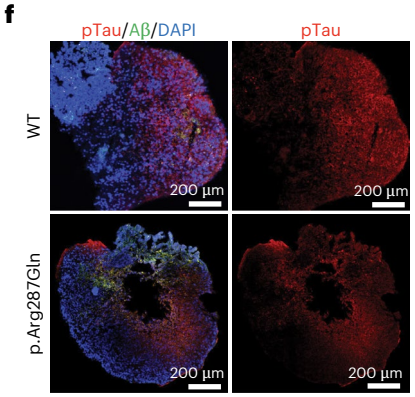


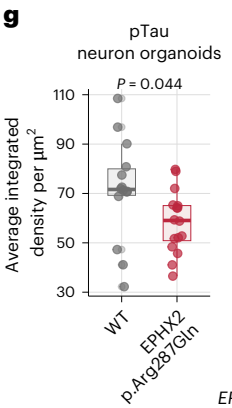


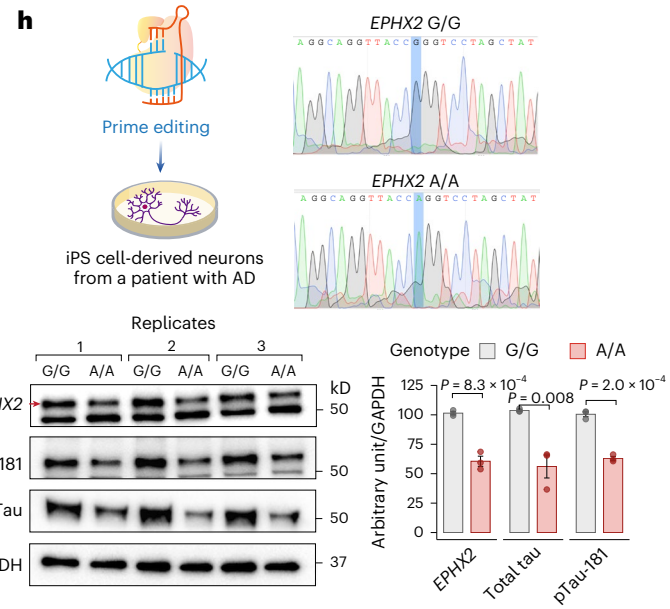


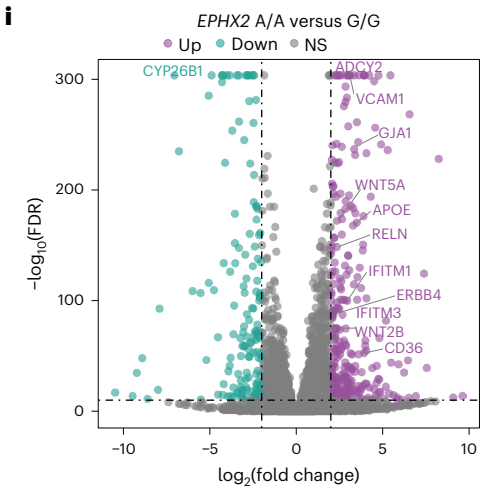


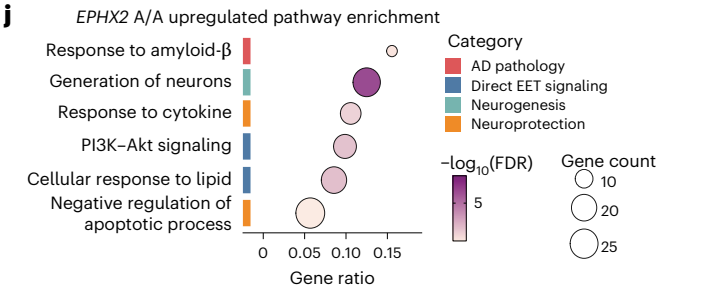












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Fig. 4 | Human genetic and functional evidence support EPHX2 as a potent anti-inflammatory target in AD. a, Two-sample MR estimates of the likely causal effect of EPHX2 on AD risk[21] using ROSMAP cortex pQTL (IVs; $n = 7$). Four MR methods were applied: IVW, maximum likelihood (MaxLik), weighted median and MR-PRESSO. Diamonds denote the β_{MR} point estimate (center), and horizontal whiskers denote the 95% CI; the dashed line marks the null ($\beta = 0$). *P* values test the null of no causal effect ($\beta_{MR} = 0$) and are two-sided. **b**, Regional plot showing genome-wide significant variants[61] associated with AD risk at *CLU* locus. Gradient color indicates pQTL *P* values of SNPs. Key variants rs1532278 (*CLU*), rs2741342 and rs751141 (p.Arg287Gln and *EPHX2*) are highlighted. **c**, CRISPRi on the non-coding SNPs rs1532278 and rs2741342 was used to test *EPHX2* and *CLU* expression in healthy iPS cell-derived neurons, using two independent guide RNA experiments with three replicates. Box plots show the median (center line), 25th–75th percentiles (box bounds) and whiskers extending to 1.5 × the interquartile range (IQR); dots are individual measurements. **d**, Western blotting of pTau levels in iPS cell-derived neurons from patients with AD, overexpressing 3xFlag–APEX2-tagged EPHX2 wild type or the p.Arg287Gln mutant. **e**, Normalized pTau-181 (top) and pTau-181/total Tau ratio (bottom) in wild-type and Arg287Gln (R287Q) samples. Data are mean ± s.e.m. with individual points ($n = 3$). **f**,**g**, Immunostaining (**f**) and quantification (**g**) showing *EPHX2*–p.Arg287Gln overexpression reduces pTau levels in iPS cell-derived cerebral organoids from patients with AD (wild-type, $n = 14$; p.Arg287Gln, $n = 16$). Box-plot elements as in **c**. **h**, Prime editing confirms that the *EPHX2* A/A genotype reduces EPHX2 protein levels compared to the G/G genotype. Bar plot shows mean ± s.e.m. with individual data points ($n = 3$). **i**, Volcano plot of differential gene expression analysis comparing *EPHX2* A/A versus G/G genotypes. **j**, Pathway enrichment analysis suggests that the *EPHX2* A/A variant activates neuroprotection, neurogenesis and anti-inflammatory pathways. A one-sided hypergeometric test was used for enrichment analysis with FDR correction for multiple comparisons. *P* values for **c**, **e**, **g** and **h** are from two-sided Student's *t*-test. MR-PRESSO: Mendelian randomization pleiotropy residual sum and outlier. WT, wild type.

EPHX2 enzymatic products, including DiHETrE and DiHDPE (Fig. 5j and Extended Data Fig. 6c). Importantly, metabolomic analysis from the Emory Healthy Brain Study[34] in CSF and plasma of patients with AD revealed elevated levels of 11,12-DiHETrE and 14,15-DiHETrE compared to cognitively normal controls. EC5026 treatment in 5xFAD mice reduced the level of 11,12-DiHETrE and 14,15-DiHETrE (Fig. 5j) in plasma, demonstrating that pharmacological EPHX2 inhibition reverses the aberrant epoxyeicosatrienoic acid metabolism observed in human AD. In addition, phenome-wide association studies[35] from the UK Biobank cohort[36] suggested no significant association for *EPHX2* IV across 624 other human disease phenotypes (Supplementary Fig. 14), indicating no deleterious effects associated with these two *EPHX2* IVs. In conclusion, we identified EPHX2 as a genetically indicated and experimentally validated potent anti-inflammatory target for AD.

Population-based validation of alzMR-score-prioritized drug targets

We conducted DWASs to further test MR-supported repurposable drugs for AD from 111,680 patients with MCI (Supplementary Table 12; Methods). To maintain statistical power, we evaluated 210 frequently prescribed drugs using four modified time-to-event regression complementary models: (1) exposed versus non-exposed model (EXP); (2) absolute higher exposed versus low exposed (Abs); (3) relatively high exposed versus low exposed (Rel); and (4) medication possession ratio (MPR). We adjusted all analyses for age, sex, race and common AD-related comorbidities and found that the percentage of significant drugs (FDR < 0.05 and hazard ratio (HR) < 1) was 7% ($n = 14$) by MPR, 9.5% ($n = 20$) by Abs, 19% ($n = 40$) by Rel and 20 % ($n = 43$) by EXP (Fig. 6a).

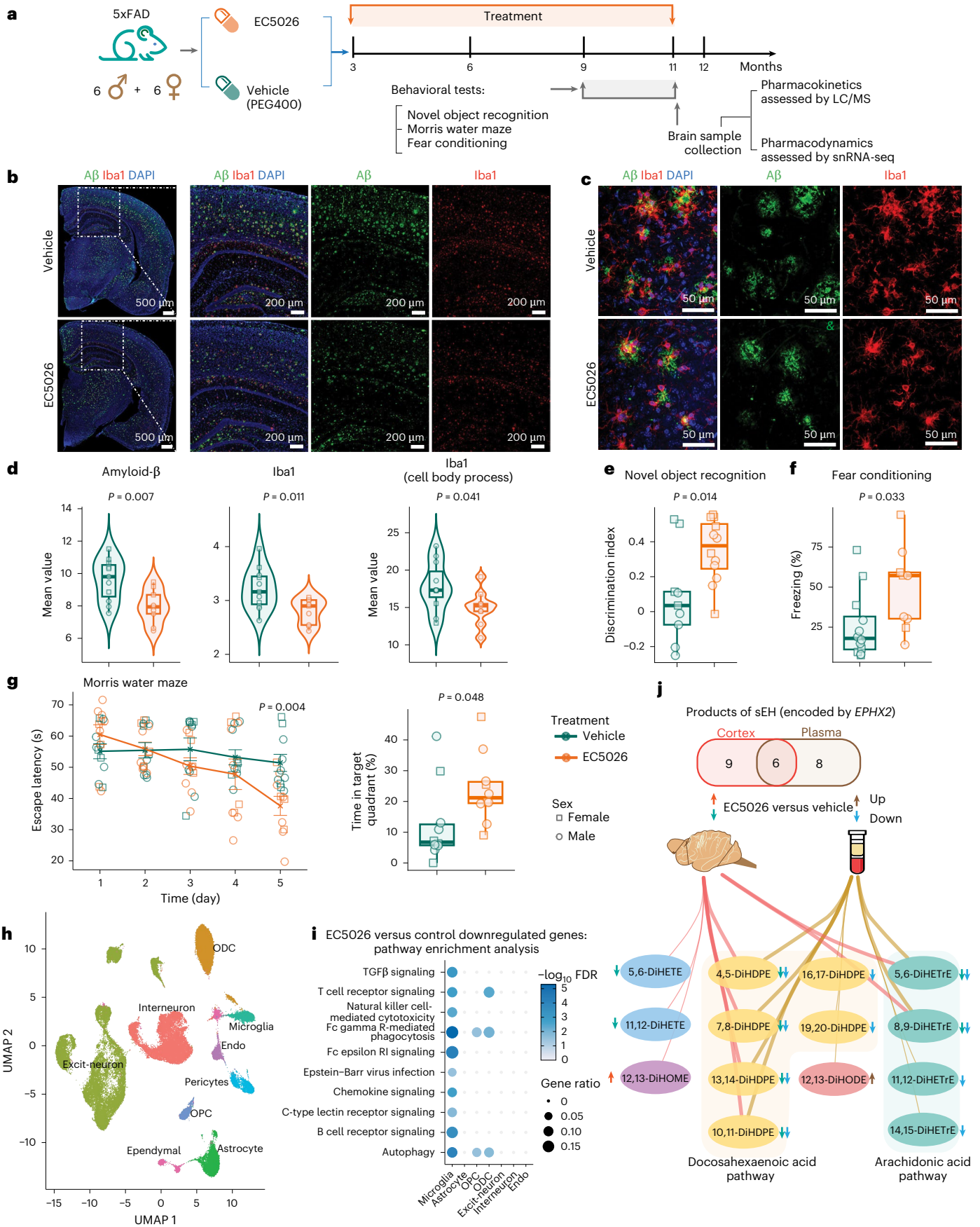
Among all significant drugs, 12 were significantly associated with reduced AD risk across all four DWAS models at an FDR < 0.05 (Fig. 6b,c). Their targets also showed significant alzMR score (Figs. 2 and 6b,c). Six drugs (50%) had significant alzMR scores (FDR < 0.05) on their primary targets, such as trazodone (ADRA1A inhibitor), losartan (acting on ACE), baclofen (GABBR1 agonist), fluticasone and spironolactone (NR3C1 agonist) and amlodipine (CACNA1D inhibitor) (Fig. 6c). Specifically, amlodipine (CACNA1D inhibitor (alzMR score = 11.4)) was significantly associated with 12% reduced risk of AD in patients with MCI (HR = 0.88; 95% confidence interval (CI) 0.85–0.91; FDR = 3.6×10^{-10}; Extended Data Fig. 7). Trazodone (ADRA1A antagonist; alzMR score = 35.7) is significantly associated with 14% reduced risk of AD in patients with MCI (HR = 0.86; 95% CI 0.80–0.91; FDR = 2.6×10^{-4}) in a dose-dependent manner (Fig. 6d and Supplementary Table 13). Additionally, trazodone was significantly associated with 22% reduced AD risk compared to an active comparator drug of escitalopram from another independent MarketScan Claim database (Methods; HR = 0.78; $P = 6.44 \times 10^{-6}$; Fig. 6d and Supplementary Table 14). In summary, these real-world patient data analyses further support alzMR-score-prioritized targets.

AD target discovery for individuals with AA

Individuals with AA and EA differ in disease prevalence, risk factors and clinical course of AD[37]. Fewer clinical trials have enrolled AA, and race-specific drug responses for AD are largely unknown[38]. We conducted drug–target MR analysis in AA (Figs. 2 and 7). We identified seven suggestive druggable targets that showed consistent direction of alzMR scores with the experimentally reported MoA of corresponding drugs ($P < 0.05$; Fig. 7, Supplementary Table 8 and Supplementary Fig. 15). ALDH2 showed consistently significant MR results in an AA-specific eQTL dataset from MetaBrain[15] (Supplementary Fig. 16). The MoAs of DHFR, SLC7A11, ALDH2 and PPARG (57%) reside in metabolism pathways (Fig. 7b and Supplementary Fig. 7b). The major function of PPARG is to regulate fatty acid storage and glucose metabolism[39], and elevated blood glucose increases the risk of dementia in AA, but not EA, with diabetes[40].

Fig. 5 | EPHX2 inhibitor EC5026 treatment reduces amyloid-β levels and neuroinflammation in 5xFAD mouse model of AD. a, Experimental design for EC5026 treatment in 5xFAD mice. We designed 12 mice per group (six female and six male) for the mouse experiments. **b**, Immunofluorescence images showing reduced amyloid-β plaques and Iba1$^+$ microglia in EC5026-treated versus vehicle-treated 5xFAD mice. **c**, Enlarged immunofluorescence images showing microglial cell body and processes surrounding amyloid-β plaques following EC5026 treatment. **d**, Quantification showing that the EC5026 treatment significantly reduced amyloid-β load, Iba1$^+$ microglial density and Iba1$^+$ microglial cell body process complexity. In total, 20 mouse samples were tested (EC5026: $n = 9$; vehicle: $n = 11$). Violin plots show the kernel density of the data, in which the width is proportional to the relative frequency of values. The box plot marks the median (center line), the 25th and 75th percentiles (box bounds) and the smallest and largest values within 1.5 × IQR of the box bounds (whiskers). **e**,**f**, EC5026-treated mice showed significantly improved cognitive performance in novel object recognition (**e**) and fear conditioning (**f**) tests. **g**, Morris water maze tests escape latency in 5 days and spent more time in the target quadrant during probe trial. Box-plot elements from **e**–**g** as in **d**. *P* values for **d**–**g** are all from two-sided Student's *t*-test. **h**, Uniform manifold approximation and projection clustering of brain cell types in EC5026-treated mouse brain samples using single-nucleus RNA sequencing (snRNA-seq) analyses (EC5026: $n = 3$; vehicle: $n = 3$). **i**, Pathway enrichment analysis of downregulated genes of EC5026 treatment. **j**, Liquid chromatography–mass spectrometry analysis of sEH metabolites in cortex and plasma (EC5026: $n = 9$; vehicle: $n = 12$), showing relative abundance of diols (DiHETrEs and DiHDPEs). DiHDPE, dihydroxydocosapentaenoic acid; DiHETrE, dihydroxyeicosatrienoic acid; Endo, endothelial cells; Excit-neurons, excitatory neurons; LC/MS, liquid chromatography–mass spectrometry; ODC, oligodendrocyte; OPC, oligodendrocyte precursor cell; TGFβ, transforming growth factor-β; UMAP, uniform manifold approximation and projection.

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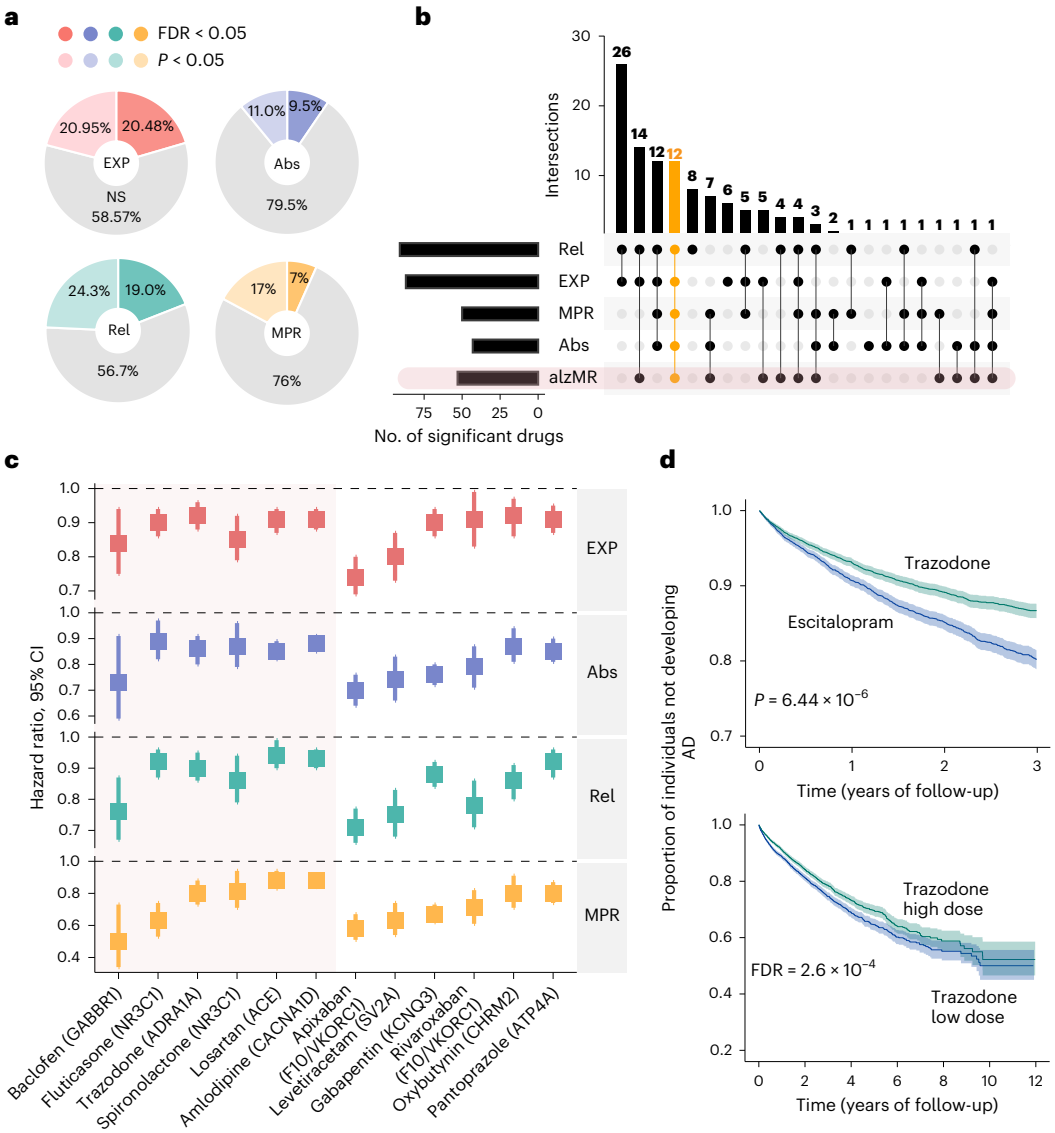


Fig. 6 | A repurposable drug compendium for AD, identified through DWASs and MR analyses. a, The percentage of significant drugs was evaluated by four pharmacoepidemiologic methods used in DWAS: exposed versus non-exposed (EXP), higher versus low absolute exposure (Abs), high versus low relative exposure (Rel) and MPR. **b**, The upset plot showing the intersections of significant drugs across the four DWAS models and MR analysis. **c**, DWAS identified 12 potential repurposable drugs for AD. HRs for AD were estimated using Cox proportional hazards regression across 111,680 patients with MCI. Squares show the HR point estimate and vertical whiskers show the 95% CI. The light-red background highlights targets supported by MR evidence at an FDR < 0.05. **d**, The curve displays the proportion of individuals who did not develop AD in the MCI population. Top: trazodone versus the comparator drug escitalopram in a target trial emulation analysis (n = 59,920). Bottom: high-dose trazodone versus low-dose trazodone exposure groups (n = 11,532). Non-exposure drug cohorts were matched to the exposures by adjusting age, sex, race, disease comorbidities and other confounding factors (Methods). Solid lines show the estimated AD-free probability; shaded bands indicate the 95% CI. P values were calculated using a two-sided log-rank test.

For AA MR analysis, *TRPV3* exhibited a stronger alzMR score (20.4; $P < 0.05$), and we detected elevated expression of *TRPV3* in the human brain in the Genotype-Tissue Expression[41] database (Supplementary Fig. 17). *TRPV3* expression in the human cortex was significantly upregulated in the late Braak stages (V–VI) compared to the early Braak stages ($P = 0.019$) and mid-Braak stages ($P = 0.026$; Fig. 7d). Elevated *TRPV3* expression was also observed across the severity of clinical cognitive diagnosis score (Supplementary Fig. 18). Dyclonine[42], a preclinical TRPV3 inhibitor (Fig. 7c), has been reported to help mitigate mitochondrial dysfunction[43].

Dihydrofolate reductase (encoded by *DHFR*) displayed the strongest alzMR score for AA (Fig. 7b). We found that *DHFR* expression in the human cortex was significantly increased in the late Braak stages compared to the early Braak stages ($P = 1.04 \times 10^{-4}$) and mid-Braak stages ($P = 0.027$) (Fig. 7d). Elevated *DHFR* expression was also observed across the severity of cognitive impairment (Supplementary Fig. 18) compared to MCI ($P = 0.003$) and non-cognitive impairment groups ($P = 1.35 \times 10^{-4}$). Real-world patient data analysis suggests that methotrexate, an FDA-approved DHFR inhibitor, was significantly associated with reduced AD risk in AA, but not EA, with MCI (Fig. 7c,e and Supplementary Table 15) in a dose-dependent manner (Fig. 7e). In summary, these MR observations suggest AA-specific candidate targets validated by real-world patient data.

Discussion

Here we report the first AD drug target compendium in EA and AA populations (Fig. 1) through large-scale genetic analysis across seven AD GWAS datasets. We identified 19 potential drug targets in EA (Fig. 3)

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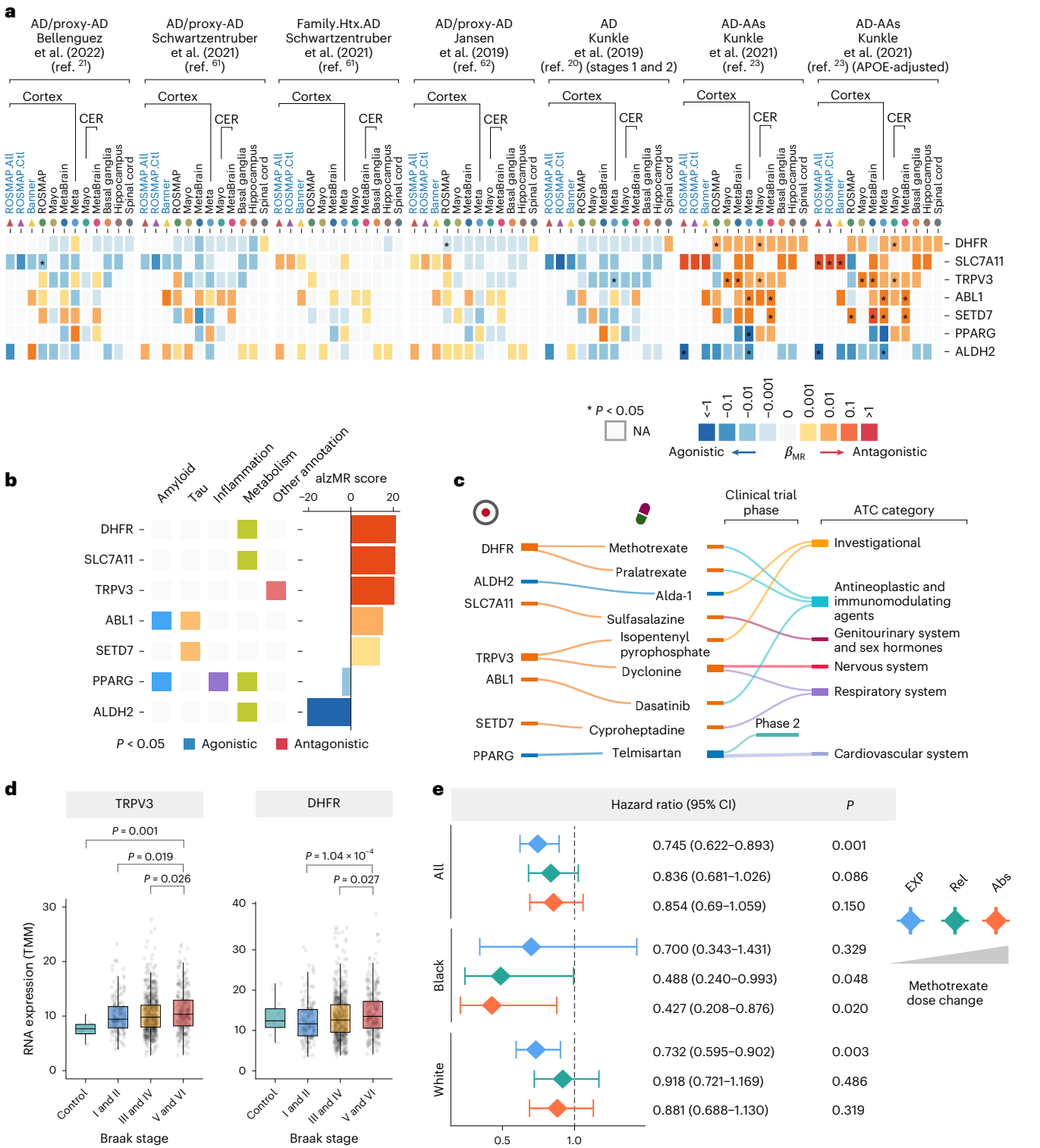


Fig. 7 | Support of human genetic evidence for AD drug targets in AA population. a, Heat map showing the β_{MR} for seven suggestive drug targets for AA[20,21,23,60–62]. β_{MR} coefficient > 0 indicates antagonistic targets; β coefficient < 0 indicates agonistic targets. NA, not available for MR analysis due to the lack of valid instrumental variables. **b**, Distribution of AA-specific drug targets across MoA categories. Left: five MoA categories of AD: amyloid, tau, inflammation, metabolism and others. Right: bar plot of alzMR score for each drug target. **c**, Sankey plot shows the relationship between repurposed drugs and alzMR score-predicted targets for AD. The dark blue color denotes that the drug is an agonist for the target; the red color denotes that the drug is an antagonist/inhibitor for the target. **d**, The relationship of *TRPV3* and *DHFR* expression with four Braak stages (control: $n = 12$; Braak I–II: $n = 173$; Braak III–IV: $n = 635$; Braak V–VI: $n = 287$). Box-plot elements similar to those presented in Fig. 4c; dots are RNA expression level for individuals. Two-sided Mann–Whitney *U*-test was performed for statistical analysis at $P < 0.05$. TMM, trimmed mean of M-values normalization. **e**, HRs indicate methotrexate dose-dependent effect in AD with AA. HRs were estimated using Cox proportional hazards regression under three DWAS exposure models with gradually increasing drug dose (Methods): exposed versus non-exposed model (EXP; all; $n = 2{,}402$; black $n = 246$; white $n = 1{,}700$), high versus low relative exposure (Rel; all; $n = 1{,}800$; black $n = 190$; white $n = 1{,}286$) and higher versus low absolute exposure (Abs; all; $n = 1{,}674$; black $n = 176$; white $n = 1{,}200$). Diamonds show the HR point estimate; horizontal whiskers show the 95% CI; dashed line marks the null (HR = 1). *P* values are two-sided, from a Wald test of log-HR in the Cox model.

and 7 in AA (Fig. 7). Our MR analysis reproducibly captured established anti-amyloid targets (BACE2 and APH1B) and neuromodulatory targets (CHRNE and GABBR1) (Fig. 3b). Consistent with recent findings[12], PDE5A emerged as a promising AD target (Fig. 3a,b). The PDE5 inhibitor AR1001 is currently being evaluated in a phase III trial (NCT05531526) in individuals with MCI or mild AD dementia defined as stages 3–4[44] (Fig. 3c). The identified neuromodulation targets were predominantly neurotransmitter receptors, including GABBR1, CHRNE and DRD2[45]. Notably, baclofen (GABBR1 agonist) was significantly associated with reduced AD incidence in individuals with MCI (Fig. 6c), supporting its future clinical evaluation.

EPHX2 encodes sEH, which hydrolyzes arachidonic-acid-derived epoxyeicosatrienoic acids and linoleic-acid-derived epoxyoctadecenoic acids to their respective diols, dihydroxyeicosatrienoic acids and dihydroxyoctadecenoic acids[46]. We identified EPHX2 as an AD drug target supported by genetic evidence across multiple GWAS datasets (Fig. 4a), in which elevated EPHX2 protein levels were associated with increased AD risk. The genome-wide significant loss-of-function variant rs751141 (p.Arg287Gln) reduces sEH protein levels ($P_{pQTL} = 5.50 \times 10^{-16}$) and alters its binding affinity for TPPU (Fig. 4b and Supplementary Fig. 11c). A second pQTL variant, rs2741342 ($P_{pQTL} = 1.19 \times 10^{-16}$), resides in an *EPHX2* enhancer region and maps to the GWAS-assigned AD risk locus for *CLU* (Fig. 4b,c). Human liver *EPHX2* pQTL[47] and eQTL[48,49] signals at rs2741342 likewise yielded significant MR evidence for AD (Supplementary Fig. 19 and Supplementary Table 16), consistent with a recent study[50]. We further demonstrated that EC5026, a picomolar sEH inhibitor, protects against cognitive impairment in 5xFAD mice (Fig. 5). Both EC5026 (NCT04228302) and AR-9281/UC1153 (NCT00847899) have established safety profiles and represent compelling candidates for AD trials.

Beyond EPHX2, we identified a set of targets at the intersection of neuroinflammation, metabolism and vascular function (for example, PLG, Fig. 3b). FOLH1, CD38 and PARP10 influence inflammatory and metabolic processes[51–53], representing potential AD targets distinct from the classical amyloid pathway, whereas PLG bridges inflammation and vascular biology through its essential roles in fibrinolysis[54] and the regulation of amyloid-β-associated brain inflammation. Collectively, these MR-supported targets provide a rich and diverse foundation for future anti-inflammatory and metabolic drug development.

Advancing ancestry-specific precision medicine is an emerging imperative in AD pharmacogenetics[55]. Among the seven AA-specific drug targets identified, DHFR and TRPV3 were the strongest candidates, each with significant alzMR scores and significant correlation of brain RNA expression with Braak stage and cognitive score (Fig. 7d and Supplementary Fig. 18). DHFR, PPARG and ALDH2 converge on metabolic pathways dysregulated in AD (Fig. 7b). Telmisartan, an FDA-approved PPARG agonist previously identified as a candidate AD treatment in AA individuals[13], was further validated by MR analysis using an AA-specific AD GWAS dataset[23]. In this study, methotrexate (an FDA-approved DHFR inhibitor) was significantly associated with reduced AD risk in AA individuals (Fig. 7e), providing independent pharmacoepidemiologic support for the DHFR signal.

Several potential limitations of this study warrant consideration. First, substantial subject overlap across the EA AD GWAS cohorts introduces non-independence among MR tests. We addressed this by applying the Cauchy combination test[56] and demonstrated that alzMR scores are not vulnerable to this inflation and reliably prioritize AD-associated genes (Supplementary Methods). Our identified drug targets are unlikely to represent a one-size-fits-all solution. Beyond ancestry differences, individual genetic variation, including *APOE* genotype, *TREM2* variants and the *EPHX2* rs751141 protective allele identified here, may substantially modulate drug response and must be incorporated in patient stratification strategies. The importance of such stratification is powerfully illustrated by the markedly higher rates of amyloid-related imaging abnormalities in *APOE* ε4 homozygous carriers treated with lecanemab[57] and by proteomic evidence that *APOE* ε4 homozygous carriers have distinct molecular profiles that could inform treatment selection[58]. Analogously, *EPHX2* rs751141 could serve as a companion biomarker to identify patients most likely to benefit from sEH inhibitors, such as EC5026. Additionally, AD diagnoses derived from real-world patient data are rarely confirmed by positron emission tomography imaging or CSF biomarkers. Future studies should prioritize biomarker-validated cohorts and apply subgroup analyses to more rigorously validate drug–disease relationships. To advance ancestry-specific precision medicine, we will apply the multiple ancestry genetic data and electronic health data from the All of Us Research Program[59], which includes 45% racial and ethnic minorities. We believe that ancestrally oriented research is scientifically and medically essential, particularly for populations currently underserved by the statistical power of existing genetic datasets.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41593-026-02472-0.

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Methods

Genetic and genomic data collection and analysis

We collected AD summary statistic datasets from seven case–control GWAS studies, including five AD GWAS datasets from EA and two AD GWAS datasets from AA[23] (Supplementary Table 2): (1) AD Bellenguez et al. (2022)[21] (GWAS catalog ID: GCST90027158) is a dataset that includes 85,934 clinically diagnosed or proxy AD cases and 401,577 controls. (2) LOAD.NoUKB23 Wightman et al. (2021)[60] is a GWAS study for late-onset AD (LOAD). In this study, data from UK Biobank and 23andMe (UKB23) were excluded from the GWAS summary statistics, in accordance with the human-subjects data-use policies of UK Biobank and 23andMe. The final cohort included 39,918 LOAD cases and 358,140 controls. (3) AD Schwartzentruber et al. (2021)[61] (GWAS catalog ID: GCST90012877) includes 75,024 cases diagnosed with AD or proxy AD, and 397,844 controls. (4) Family.Htx.AD Schwartzentruber et al. (2021)[61] (GWAS catalog ID: GCST90012878) includes 53,042 family history of AD cases and 355,900 controls. (5) AD Jansen et al. (2019)[62] (GWAS catalog ID: GCST007320) includes 71,880 cases of LOAD and proxy AD, and 383,378 controls. (6) Kunkle et al. GWAS data (2019)[20] include both stage 1 and stage 2 datasets. The stage 2 dataset genotyped the 11,540 variants on the I-select chip for an independent cohort (8,362 clinically diagnosed AD cases and 10,483 controls). Stage 2 also includes the meta-analysis results from both stage 1 and stage 2 (30,344 cases and 52,427 controls). (7) AD-AA Kunkle et al. (2021)[23] model1 (M1), and AD-AA Kunkle (2020) model2 (M2) were accessed from the National Institute on Aging Genetics of Alzheimer's Disease Data Storage, including 2,784 AD cases and 5,222 cognitively normal controls. These two AA datasets used the same cohort under different covariate-adjustment models: M1 conducted a meta-analysis with adjustment of age and sex, whereas the meta-analysis results of the M2 adjusted age, sex and APOE genotypes.

Drug–target interactions

We incorporated drug–target interaction data from multiple sources, including BindingDB[63], ChEMBL[64] and GtoPdb[65] (December 2021). We first extracted the drugs from DrugBank[66] and mapped molecules from these three databases to the drugs using available information, including names, simplified molecular input line entry system or other database-specific identifiers. Ambiguously mapped data were removed. We mapped human proteins to their Entrez Gene IDs and included only physical interactions from these data sources. We used a cutoff of inhibition constant/potency (K_i), dissociation constant (K_d), median effective concentration (EC_{50}) or median inhibitory concentration (IC_{50}) ≤ 1 μM to filter the interactions. The final drug–target interaction network with activity values for each interaction contained 13,154 interactions among 3,079 drugs and 1,430 genes/proteins. Among them, 1,229 genes/proteins have valid IVs from eQTL or pQTL datasets and were used in subsequent MR analysis.

Instrument sources

In this study, we utilized three pQTL and nine eQTL datasets of human brain samples and cell-type-specific eQTL datasets from multiple biobanks (Supplementary Table 1). Of the three pQTL datasets of human cortex samples, two were generated from ROSMAP[14]: (1) a pQTL dataset with AD and control participants and (2) a pQTL dataset with control participants only; (3) a third pQTL dataset was from the Banner Brain Projects. The following nine *cis*-eQTL instrument sources are publicly available from the AD Knowledge Portal[67] (Synapse ID: syn17015233) and MetaBrain[68]: three eQTL datasets of cortex samples generated by (1) ROSMAP[69–71]; (2) Mayo Clinic BiobanK[16]; and (3) MetaBrain[68]; two eQTL datasets from cerebrum samples generated from (4) Mayo Clinic[16] and (5) MetaBrain[68]; (6) a meta-analysis eQTL dataset combined with datasets[67] 1, 2 and 4; (7) an eQTL dataset of basal ganglia samples[68]; (8) an eQTL dataset of hippocampus samples[68]; and (9) an eQTL dataset of spinal cord samples[68] collected from MetaBrain. Cell-type-specific eQTL datasets were also included: (1) four microglia-specific eQTL datasets published by Lopes et al.[72]; (2) seven cell-type-specific eQTL datasets published by Bryois et al.[73]; and (3) six cell-type-specific eQTL datasets published by Fujita et al.[74].

MR analysis

The MR[75,76] analysis was conducted to test the casual effect of 1,229 drug targets for AD (Extended Data Fig. 1). A two-sample polygenic MR approach was applied[77]. We used four criteria to select high confidence IVs[78]: (1) FDR < 0.05 was the cutoff for actionable IVs from eQTL and pQTL datasets; (2) *F*-statistics > 10 was used as the cutoff to remove weak instruments, ensuring that the pQTL/eQTL IVs were robustly associated with the drug targets of interest; (3) LD pruning at $r^2 < 0.2$ based on reference matrices from 1000 Genomes V3 using PLINK 1.9 (ref. 79), in which SNPs in LD within 500-kb window were pruned; and (4) to strictly satisfy the independence assumption of MR, IVs associated with more than two proteins or RNA transcripts across all QTL datasets were excluded.

Next, we designed a framework for MR model selection depending on the number of proposed IVs for the drug target. In total, five MR models were conducted for all drug–target validations: (1) For IV = 1, the MR method Wald ratio estimator[80] was used for the instrument with one variant in all drug targets. (2) For IV = 2, the IVW fixed-effect model[17,76] was used to test drug targets with two IVs. (3) For IV ≥ 3, four MR models were used to evaluate the drug targets, including IVW random-effect model, MR-PRESSO[18], MaxLik and weighted median. (4) For drug targets with three or more IVs, heterogeneity was tested using the IVW and MaxLik models. (5) When Cochran's Q $P < 0.05$ and $I^2 > 50\%$, we applied robust IVW and MaxLik models that down-weighted outlier IVs. MR analysis was conducted using the TwoSampleMR (https://mrcieu.github.io/TwoSampleMR/) and MendelianRandomization[81] packages in R (v.4.1.1) platform.

AlzMR score

The alzMR score is intended to be largest for genes with the strongest causal effects and for which we have the highest degree of confidence for AD (Supplementary Methods):

$$\text{alzMR}(g) = \sum_j \sum_k Z_{jkg}$$

Here Z_{jkg} is the Z-statistics used to test the causal null hypothesis in the kth xQTL GWAS cohort and jth AD GWAS cohort for gene g. The alzMR score for gene g (alzMR) is the summation of Z-statistics across all $k = 1, \ldots, K$ and $j = 1, \ldots, J$ cohorts. Because not all xQTLs are testable in MR, the alzMR score may be largest for genes with the strongest xQTLs. We also provide a normalized alzMR score ($\frac{\text{alzMR}(g)}{N}$), calculated by dividing the alzMR score introduced above by the number of pQTL/eQTL types and AD GWAS datasets used (N). The Cauchy combination test[56] was used to validate the alzMR scores, accounting for overlap across six EA AD GWAS datasets. This approach provides a single robust P value per gene, enabling consistent and interpretable results across different datasets. A drug target with a higher absolute alzMR score (|alzMR score|) indicates a higher likelihood of disease modification. An alzMR score > 0 indicates that elevated gene (eQTL) or protein (pQTL) expression of a specific drug target is associated with elevated AD risk. We therefore termed a target harboring an alzMR score > 0 as antagonistic because its antagonism or inhibition could offer potential treatment for AD. By contrast, an alzMR score < 0 indicates that elevated gene (eQTL) or protein (pQTL) expression is associated with reduced AD risk. These were defined as agonistic targets.

Colocalization analysis and fine-mapping analysis

We conducted colocalization analysis between pQTL/eQTL ($n = 12$) and AD GWAS ($n = 6$ in EA) data using the coloc package[82] in R (v.4.1.1)

platform. The lead SNPs across all loci in AD GWAS were collected from six published AD GWAS studies. Overlapping SNPs between QTL and GWAS datasets within 500-kb flanking windows were used for colocalization analysis across each locus. Loci with PP.H4 > 0.5 were considered casual genes for AD. We also performed FOCUS[19] analysis using two eQTL weight datasets (downloaded from the FUSION database, http://gusevlab.org/projects/fusion/) of the brain cortex region and seven AD GWAS datasets. Genes with PIP > 0.5 were considered causal genes for AD.

Conditional analysis and multicausal joint estimate analysis

We found that multiple lead SNPs of *CLU* have been reported across different AD GWAS cohorts. Therefore, we used all reported lead SNPs related to *CLU* and *PTK2B* as conditional SNPs to test the significance and effect sizes of *EPHX2* SNPs rs2741342 and rs751141 (Supplementary Table 10). To better understand the extent to which genetic associations in this locus are independent of each other, we estimated joint SNP associations using the conditional and joint multi-SNP method[83] with LD estimates from 1000 Genomes Phase 3 and seven AD GWAS datasets from EA (Extended Data Fig. 4b). Because conditional and joint multi-SNP conditional analysis has an inflated false-positive rate when the number of individuals in the LD reference panel is small relative to the number of tested SNPs, we applied corrected standard errors, which are shown to be unbiased multicausal joint estimate analysis (Supplementary Methods).

Drugome-wide association studies

The pharmacoepidemiology study used Optum's de-identified Clinformatics Data Mart database (2007–2020), derived from commercially insured US individuals and the US Medicare Advantage beneficiaries. We included 111,680 patients with MCI and no previous history of AD, cancer, Parkinson's disease and hemiplegia or paraplegia. We computed the MPRs for 210 drugs and used a multiple Cox regression model to test associations between MPRs and time to AD under four exposure models: (1) exposure (MPR > 0) versus no exposure (MPR = 0); (2) high exposure (MPR > 0.25) versus low exposure (MPR ≤ 0.25); (3) relative high exposure (MPR > 25% quantile of the corresponding drug MPRs) versus relative low exposure (MPR ≤ 25% quantile of the corresponding drug MPRs); and (4) using MPR directly. We used R packages MatchIt[84] and optmatch[85] to derive 1:1 propensity-score-matched datasets, as well as the Kaplan–Meier estimator and log-rank test to compare the probability of not developing AD across matched groups. The top candidate drugs were further validated in an independent cohort from the MarketScan insurance claims database (2011–2020) using weighted Kaplan–Meier estimators and inverse probability weighting to estimate cumulative incidence curves for trazodone versus the active comparator escitalopram. Full cohort definitions (Supplementary Tables 12–14), covariates and statistical models are provided in Supplementary Methods.

Functional experiments of p.Arg287Gln in *EPHX2*

iPS cell-derived neuron from a patient with AD. Forebrain neurons were differentiated from iPS cells derived from a 63-year-old female patient with LOAD (IUGB269.1), provided by the National Centralized Repository for Alzheimer's Disease and Related Dementias at Indiana University. We differentiated the iPSCs into mature forebrain neurons using a commercial pipeline from STEMCELL Technologies. iPS cells were initially cultured in mTeSR 1 medium, dissociated and plated on Matrigel-coated dishes. They were cultured in neural induction medium (05839) to form neural progenitor cells. The neural progenitor cells were later cultured in forebrain neuron differentiation (08600) and maturation (08605) media. After 10 days of maturation, the neurons were treated with 2 μg ml^{-1} doxycycline for 48 h. This process was conducted in a controlled environment of 37 °C and 5% CO_2. Finally, western blotting was conducted to test pTau-181 (1:1,000; Cell Signaling Technology; 12885), tau (1:1,000; Cell Signaling Technology; 46687), Flag (1:2,000; Sigma-Aldrich; F3165) and glyceraldehyde-3-phosphate dehydrogenase (1:2,000; Cell Signaling Technology; 97166) using a well-established pipeline[86]. ImageJ was used to quantify the protein bands from the western blot image.

Doxycycline-inducible overexpression system. We designed an experiment to functionally test the association of the *EPHX2*–p.Arg287Gln mutation with AD risk using human iPS cell-derived neurons from a patient with LOAD. We used the TALEN system[31,87] to insert a doxycycline-inducible overexpressed EPHX2 or *EPHX2*–p.Arg287Gln at the *AAVS1* locus in AD iPS cells. In brief, four vector systems were purchased from Addgene: Puro–Cas9 donor (plasmid 58409), AAVS1–Neo–M2rtTA (plasmid 60843), AAVS1–TALEN-R (plasmid 59026) and AAVS1–TALEN-L (plasmid 59025). To create EPHX2 or *EPHX2*–p.Arg287Gln mutant inducible system, the sequences of EPHX2 or *EPHX2*–p.Arg287Gln mutant with 3xFlag–APEX2 tags were subcloned into Puro–Cas9 donor plasmid to replace Cas9 sequences. iPS cells derived from a patient with AD were dissociated with Accutase and transfected with one of the EPHX2 or *EPHX2*–p.Arg287Gln donor vectors, along with AAVS1–Neo–M2rtTA and AAVS1–TALEN pairs (R and L) using a Human Stem Cell Nucleofector Kit 1 (Lonza) with the Nucleofector 2b Device (Lonza). CloneR (50888) was used to aid iPS cell recovery within 24 h, and G418 (50 μg ml^{-1}) was treated for 7–10 days. Puromycin (0.5–1 μg ml^{-1}) was added 48–72 h later to select for cells with genetic integration. After 3 weeks, we selected three EPHX2 and three *EPHX2*–p.Arg287Gln mutant colonies from the cell dish. These iPS cell colonies were then used to differentiate into neurons for further testing for AD tau pathology.

EC5026 treatment in 5xFAD mouse

Ethics statement for mouse experiments. All protocols were approved by the Cleveland Clinic Lerner Research Institute Institutional Animal Care and Use Committee in compliance with the guidelines established by the Public Health Service Guide for the Care and Use of Laboratory Animals.

The mouse strain used for this research project, B6SJL-Tg (APPSwFlLon, PSEN1*M146L*L286V) 6799Vas/Mmjax, research resource identifier: MMRRC_034840-JAX (ref. 88) was obtained from the Mutant Mouse Resource and Research Center (MMRRC) at The Jackson Laboratory, an NIH-funded strain repository, and was donated to the MMRRC by Robert Vassar, Northwestern University. Mice were purchased at 6–8 weeks of age. They were group-housed in standard vivarium conditions. The vivarium was maintained at 20–22 °C and 30–70% humidity. Lights were kept on a 12-h light/12-h dark cycle, and experiments were conducted during the light time. The mice were randomly assigned to EC5026 or vehicle treatment groups. EC5026 was dissolved in 1% polyethylene glycol, molecular weight 400 (PEG-400; Thermo Fisher Scientific) for in vivo treatment. 5xFAD transgenic mice received EC5026 (3 mg kg^{-1} d^{-1}) in drinking water or vehicle (1% PEG-400) for 8 months starting at 3 months of age. Both sexes were included in all experiments, with six female and six male mice per group at the start. The final sample size analyzed varied across experiments. Animals were excluded if they died, escaped from the testing chamber (fear conditioning), exhibited immobility or non-swimming behavior (Morris water maze) or explored the objects for <25 s total (novel object recognition). These exclusion criteria were applied equally across groups.

Mouse behavioral tests

Novel object recognition was done under dim white light (~30 lux). On day 1, a single 5-min habituation session was performed: mice were placed inside a Plexiglas arena box (45 × 45 × 45 cm; $L \times W \times H$) and allowed to explore freely while a video camera above the testing boxes, together with tracking software (EthoVision XT; Noldus), recorded their activity. After 5 min, the mice were removed from the box and

placed back into their home cage. On day 2 (training phase), the mice were placed in the same box with the addition of two identical objects and allowed to explore the objects freely for 10 min. On day 3 (object recognition phase), the mice were brought back into the same arena box. One of the objects was explored previously (familiar object). Another one was replaced with a new object (differing in color and shape but sharing a common size and volume). The mice were allowed to explore the objects again for 10 min. The time the mice spent exploring the objects was assessed. The discrimination index was calculated using the following formula: discrimination index = (time exploring new object -time exploring old object) / (time exploring new object + time exploring old object). For all animal experiments, investigators were blinded to the mouse treatment groups during behavioral and other testing.

Morris water maze is a standard test used to evaluate spatial learning and memory. The apparatus consists of a circular pool (1.2 m in diameter) filled with opaque water (nontoxic, water-soluble tempera white paint) maintained at a constant temperature of 24 °C using an internal pool heater, and a submerged goal platform (1 cm below the water). Testing was done in a dimly lit room (~40 lux) with spatial cues on all four walls surrounding the water maze. The training phase began by placing each mouse into the pool, facing the wall. A tracking system (EthoVision XT 15; Noldus) was set up to track each mouse once it entered the pool arena. Using the software, the pool was divided into four quadrants (target, opposite, near and far). The software tracked each mouse as it swam across these quadrants and over the hidden platform. Each mouse received four training trials per day and was introduced into the pool in a randomized sequence of locations (target, opposite, near and far) each day. Mice were allowed 60 s to swim in the pool. If a mouse was unable to find the submerged platform, the experimenter guided it to the platform. After reaching the platform and remaining on it for 2 s, the tracking software stopped the trial, and the mouse was left on the platform for 15 s. The mouse was then removed from the pool, dried off with a super-absorbent paper towel and placed in its holding cage to await another trial or returned to its home cage if all four trials were completed. Training trials were conducted for five consecutive days (days 1–5). After 24 h from the last training trial, a 60-s probe trial was conducted without the platform in the pool.

Contextual fear conditioning consisted of a training period and a testing period. During the training period, mice were placed in fear conditioning boxes consisting of a Plexiglas shock box with clear front and rear walls, a gridded floor with bedding underneath, an internal light source for both white light and near-infrared light and a fixed external near-infrared camera for video recording. The mice were placed inside the chamber for 2 min, after which they received a 2-s 0.6-mA foot shock (unconditioned stimulus). The mice received a total of three shocks, each separated by 2-min intervals. The mice remained in the chamber for 2 min after the shocks before being returned to their home cage. Between mouse trainings, the bedding was replaced. Twenty-four hours after the training period, the context test was conducted by returning the mice to the same fear conditioning box with the same contextual cues (chamber shape, lighting and scent) for 5 min without stimulus or shock. Freezing behavior was monitored, automatically recorded and analyzed using VideoFreeze software (Med Associates).

Immunohistochemistry and imaging

Organoids and mouse brains were cut into 20-µm-thick slices on a cryostat microtome (Leica; CM1860) and mounted onto charged glass slides. The sections were treated with 0.5% Triton X-100 for 30 min and blocked in 10% fetal bovine serum (FBS) for 1 h. For organoid sections, anti-amyloid (B-4) Alexa Fluor 488 conjugated antibody (1:200; Santa Cruz Biotechnology; sc-28365) and anti-tau (phospho T205) Alexa Fluor 647 conjugated antibody (1:200; Abcam; AB307376) were used to detect amyloid-β and phosphorylated tau, respectively. Antibodies were stained for 2 h at room temperature and washed three times with phosphate-buffered saline (PBS) (20 min each) before mounting in Vectashield medium with DAPI (Vector Laboratories). Mouse brain slices were treated with primary antibodies diluted in 10% FBS overnight at room temperature. Primary antibodies included rabbit anti-Iba1 (1:500; FUJIFILM Wako; 019-19741) and mouse anti-amyloid (1:500; BioLegend; 803001). After washing off excess primary antibodies three times with phosphate-buffered saline for 20 min each, sections were incubated with secondary antibodies diluted in 10% FBS at room temperature for 2 h. Secondary antibodies included donkey anti-mouse Alexa Fluor 488 (1:1,000; Thermo Fisher Scientific; A-21202) and donkey anti-rabbit Alexa Fluor 568 (1:1,000; Thermo Fisher Scientific; A10042). Finally, sections were mounted in Vectashield medium with DAPI (Vector Laboratories) for imaging. Images were acquired using an upright fluorescent microscope (Leica) or an inverted confocal microscope (Leica SP8). Objective magnification (×10 or ×63) was selected based on the desired image resolution. Serial *z*-stack images were taken when structures spanned multiple layers. In such cases, a composite image was generated by projecting at maximum intensity. Images were processed and analyzed using ImageJ (v.1.53k).

Single-nucleus transcriptome sequencing and bioinformatics analysis

The frozen mouse brain samples were sent to Azenta Life Sciences for snRNA-seq using the default 10x Genomics Chromium 3′ gene expression pipeline. Raw sequencing reads were processed using Cell Ranger (v.7.0.0; 10x Genomics) to align reads to the mouse reference genome (refdata-gex-mm10-2020-A) and generate cell-by-gene count matrices for each sample. FASTQ files were returned for quality control using FastQC and for doublet removal using DoubletFinder. Seurat (v.4.0.2) was used for downstream analysis. Uniform manifold approximation and projection was used for nucleus clustering, and cell-type annotations were based on previously published markers (Extended Data Fig. 5b). We identified seven major cell types in the mouse brain cortex (Fig. 5h), including astrocyte, microglia, interneuron, excitatory neurons, oligodendrocytes, oligodendrocyte precursor cell and endothelial cells. The MAST function was used to analyze differential gene expression between EC5026-treated and control groups. Gene-set enrichment analysis was performed using the R package clusterProfiler. All snRNA-seq analyses were conducted on the R platform (v.4.1.1).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Details of the GWAS and eQTL/pQTL datasets used in this study are provided in Supplementary Tables 1 and 2. All MR results are available via AlzTarget (https://alztarget.org/). The bulk RNA-seq data from iPS cell-derived neurons carrying the *EPHX2* mutation, generated in this study, are available in the Gene Expression Omnibus under accession number GSE339305. The mouse brain single-nucleus RNA-seq data generated in this study are available under accession number GSE339306. The bulk RNA-seq data are available from ROSMAP at syn3219045. The H3K4me3 and H3K27ac ChIP–seq data from the hippocampus are available under accession numbers GSM773022 and GSM773020. The H3K4me3 and H3K27ac ChIP–seq data from the inferior temporal lobe are available under accession numbers GSM772996 and GSM772995. The single-cell ATAC–seq and HiChIP data are available under accession number GSE147672. Source data are provided with this paper.

Code availability

The code for MR and DWAS analysis is available via GitHub at https://github.com/ChengF-Lab/alzMRTargets.

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Author contributions

F.C. conceived the study. Y.H. performed all genetic and genomic data analysis and experiments. P.Z. performed patient data analysis using the Optum Claims database. E.K. and M.D. performed patient data analysis using the MarketScan Claims database under the supervision of M.R.-Z. Y.L. and Z.T. performed all mouse experiments and data analyses. F.F., X.F. J.B., H.Z., R.J.R. and J.N.M. supported data interpretation and discussion. S.H.H. and B.D.H. provided the EC5026 compound. J.Y. conducted metabolomic analysis of mouse plasma and brain samples. D.G., X.C., W.S. and X.Y. conducted CRISPRi and prime editing experiments and data analyses. F.C., Y.H., N.L.-C., W.S., W.M., Y.F., J.X., L.D., T.B., Z.L., I.R.P., J.L.H., H.Y., Y.Q., Y.Z., L.B., A.A.P., J.B.L. and J.C. interpreted the data analyses. Y.H., F.C. and P.Z. drafted the paper. F.C., Y.H., Y.L., N.L.-C., P.Z., A.A.P., J.C. and J.B.L. critically revised the paper. All authors critically revised and approved the final version of the paper.

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Competing interests

J.C. has provided consultation to AB Science, Acadia, Alkahest, AlphaCognition, ALZPath, Annovis, AriBio, Artery, Avanir, Biogen, Biosplice, Cassava, Cerevel, Clinilabs, Cortexyme, Diadem, EIP Pharma, Eisai, GatehouseBio, GemVax, Genentech, Green Valley, Grifols, Janssen, Karuna, Lexeo, Lilly, Lundbeck, LSP, Merck, NervGen, Novo Nordisk, Oligomerix, Ono, Otsuka, PharmacotrophiX, PRODEO, Prothena, ReMYND, Renew, Resverlogix, Roche, Signant Health, Suven, Unlearn AI, Vaxxinity, Vigil Neuro pharmaceutical, assessment and investment companies. J.B.L. has received consulting fees from Vaxxinity and grant support from GE HealthCare. He serves on a Data Safety Monitoring Board for Eisai. B.D.H. is a cofounder and CEO of EicOsis LLC. S.H.H. is a part-time employee of EicOsis LLC. H.Z. serves on the Scientific Advisory Board of the Tau Consortium, which is managed by the Rainwater Charitable Foundation. E.K., M.D. and M.R.Z. are employees of the IBM Research. The other authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41593-026-02472-0.

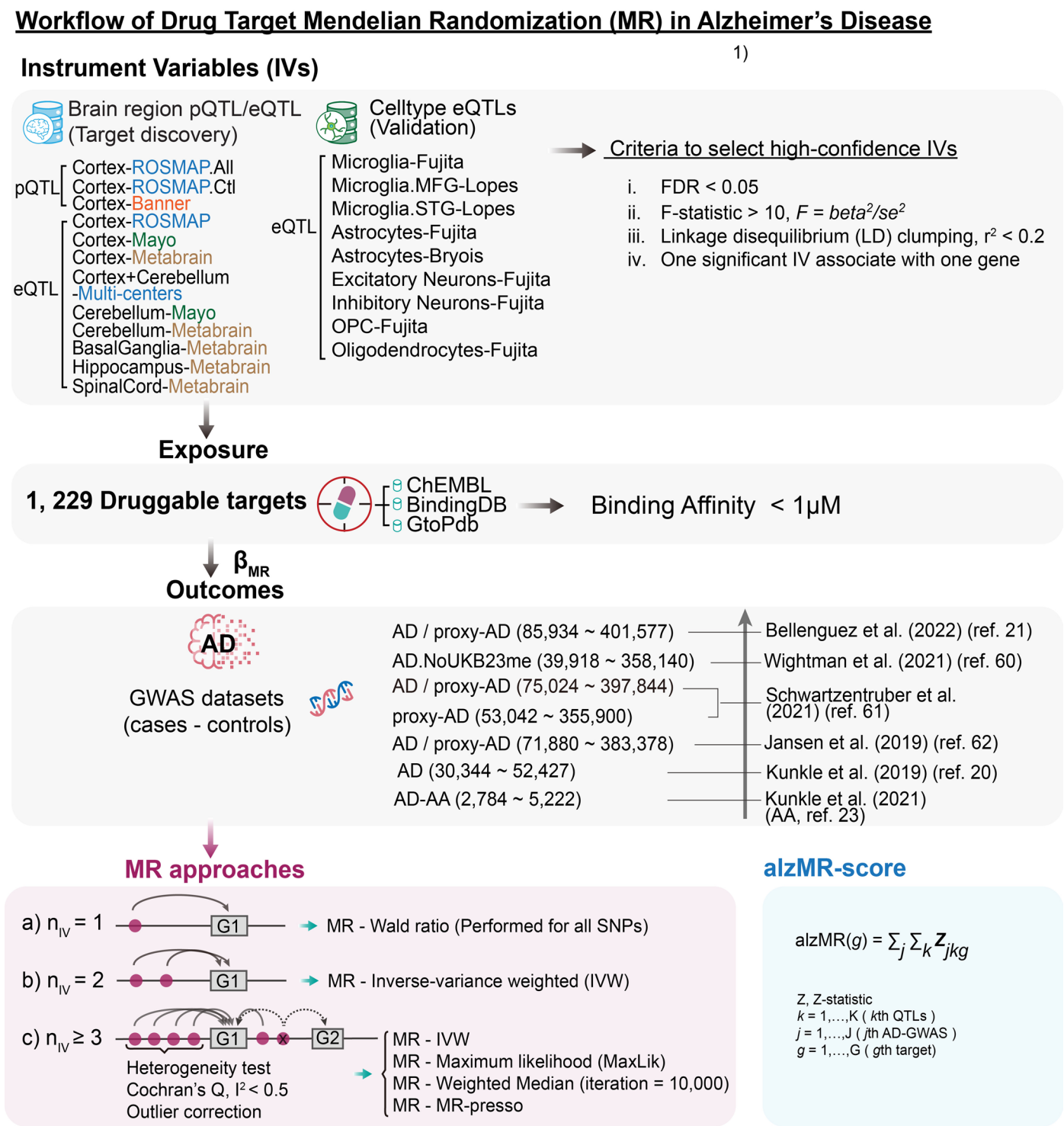
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41593-026-02472-0.

Correspondence and requests for materials should be addressed to Feixiong Cheng.

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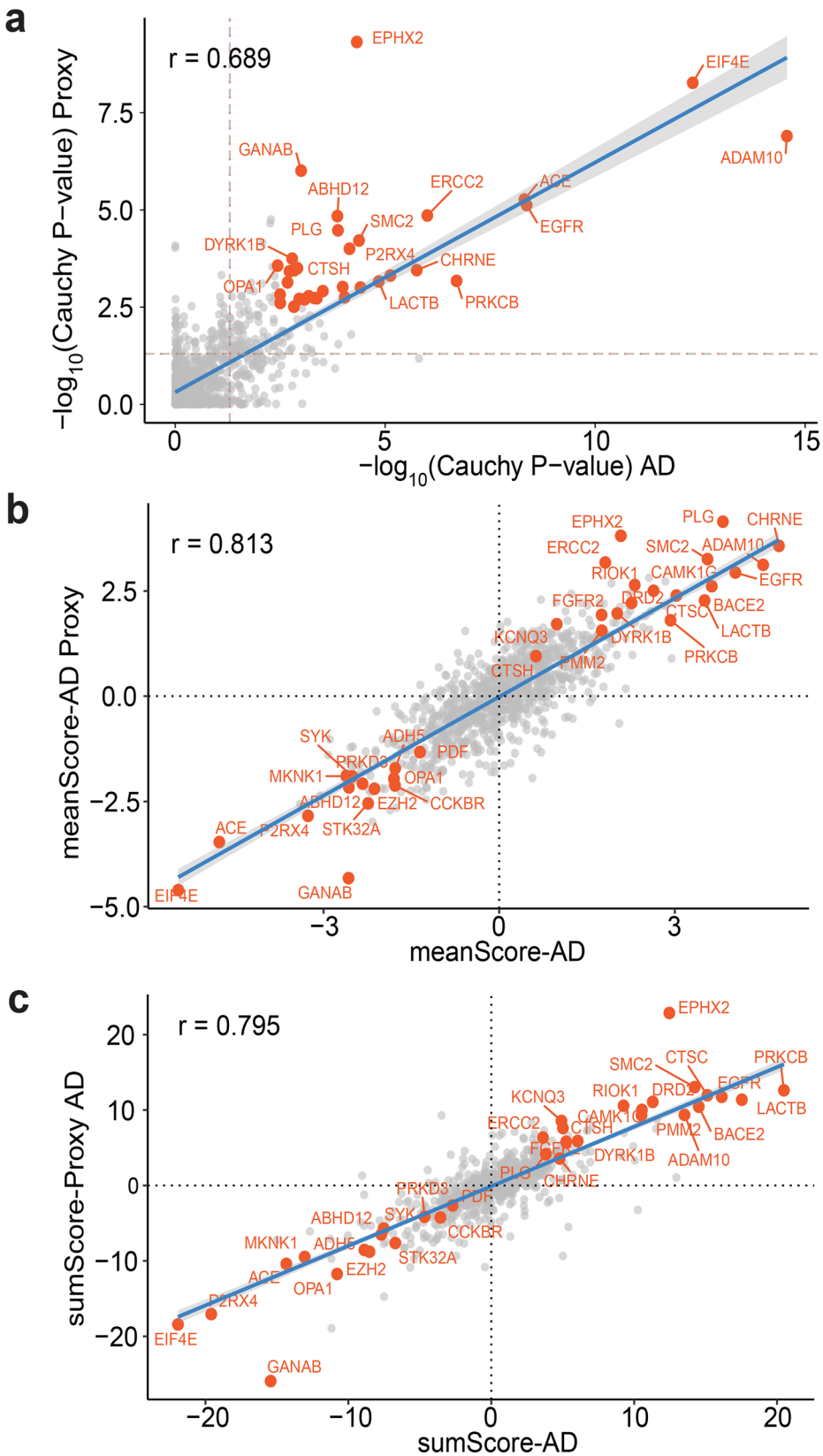
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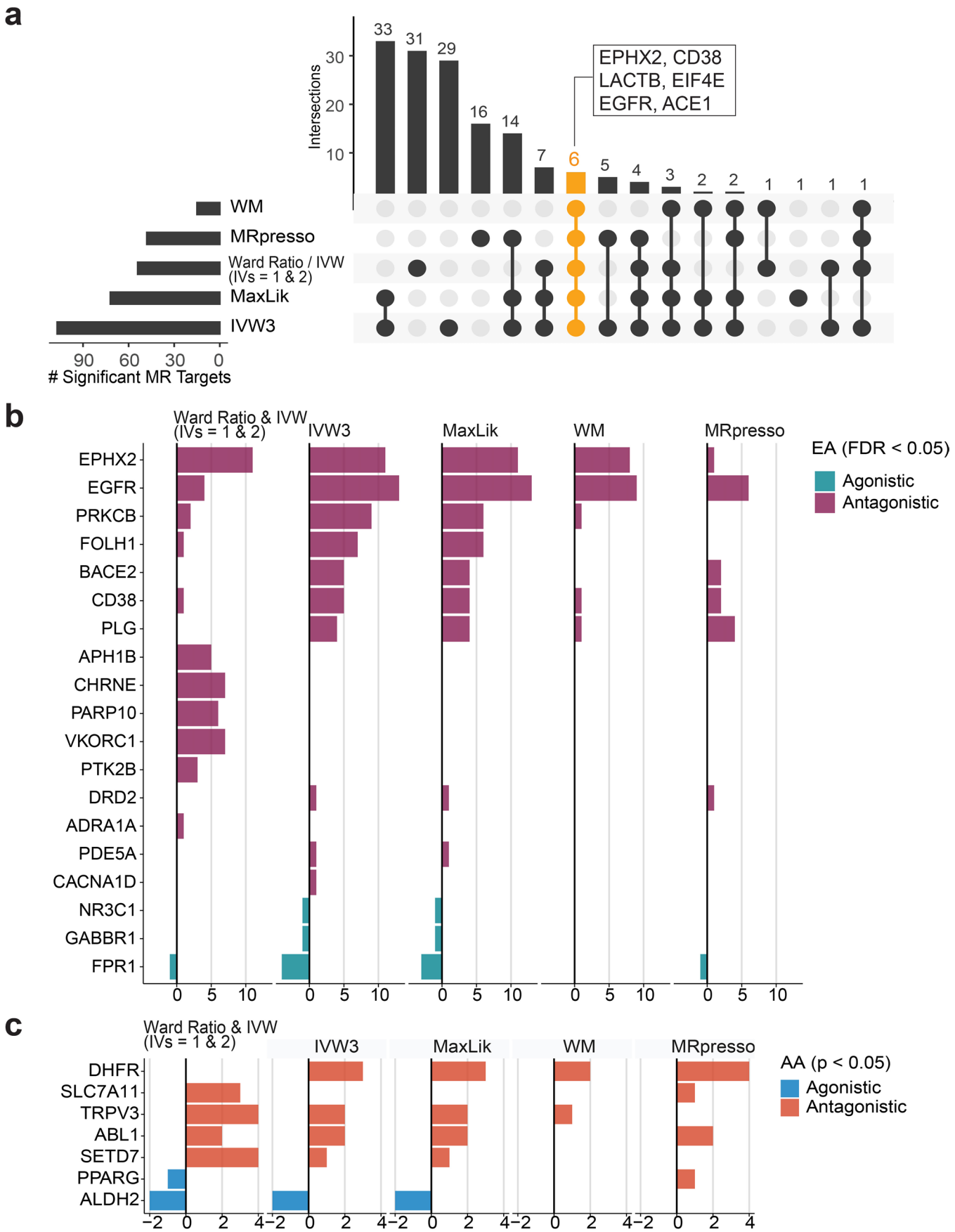
Extended Data Fig. 1 | Overview of the Mendelian randomization (MR) analysis workflow. Schematic of the drug-target discovery pipeline: selection of 1,229 druggable proteins; selection of instrumental variables (IVs; FDR-adjusted P < 0.05 and F-statistic > 10) from 3 protein quantitative trait loci (pQTL) and 9 expression quantitative trait loci (eQTL) datasets across 5 brain regions; and the five complementary MR models applied to 7 AD GWAS outcome datasets (European and African ancestry).

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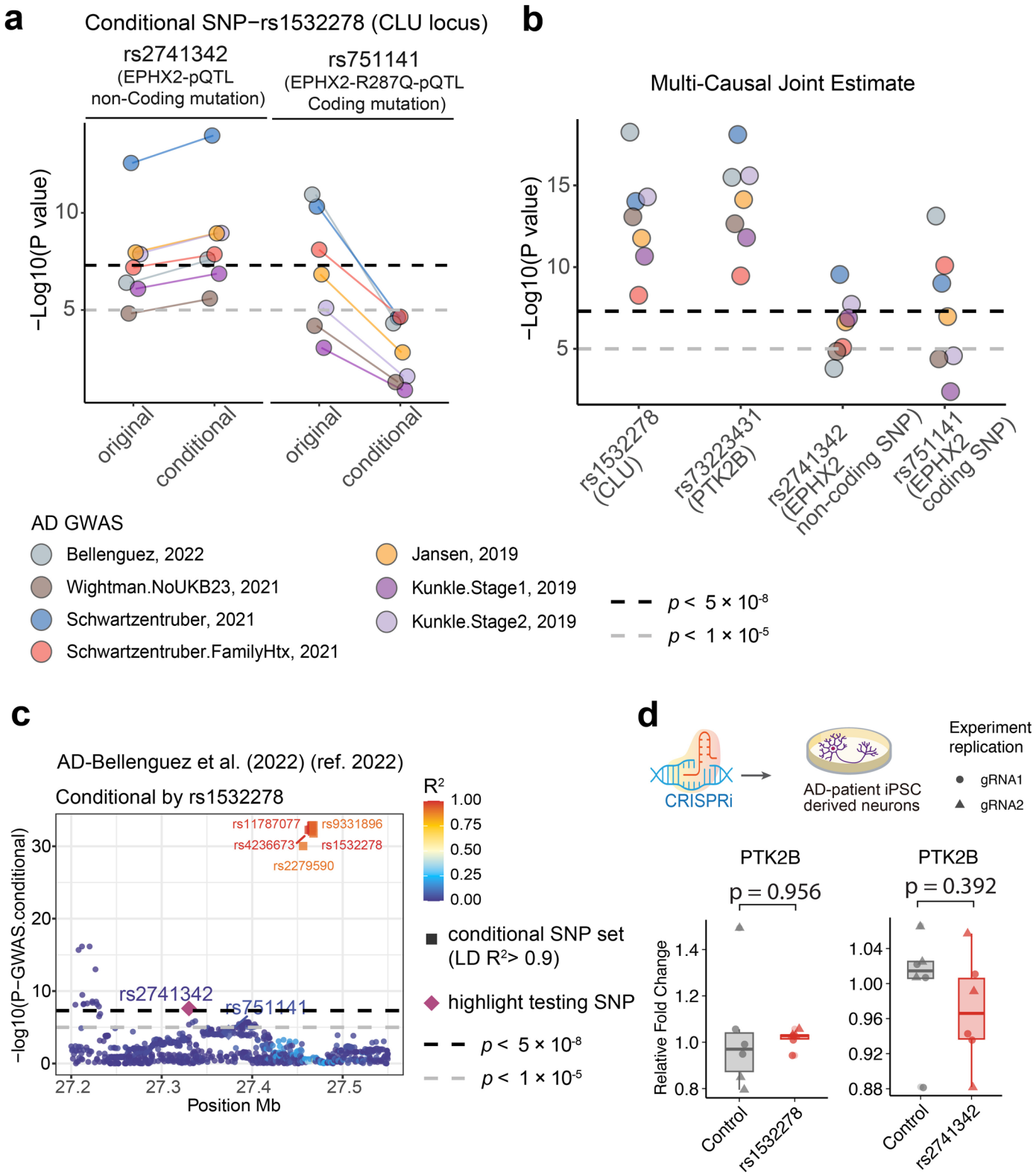
Extended Data Fig. 2 | Concordance between Cauchy combination test from AD and proxy AD GWAS cohorts. a, Association between transformed Cauchy P-values calculated in the diagnosed AD and proxy AD (y-axis, UK-biobank family history GWAS summary data only, Schwartzentruber et al. 2021, GWAS catalog ID: GCST90012878) cohorts. Cauchy combination P-values are calculated within genes across pairs of xQTL types and AD cohorts. Each point represents a single gene. **b-c**, Concordance between mean-alzMR scores (**b**) and sum-alzMR score (**c**) from AD and proxy AD GWAS cohorts. Each point is a druggable target; the correlation coefficient was computed by two-sided Pearson correlation.

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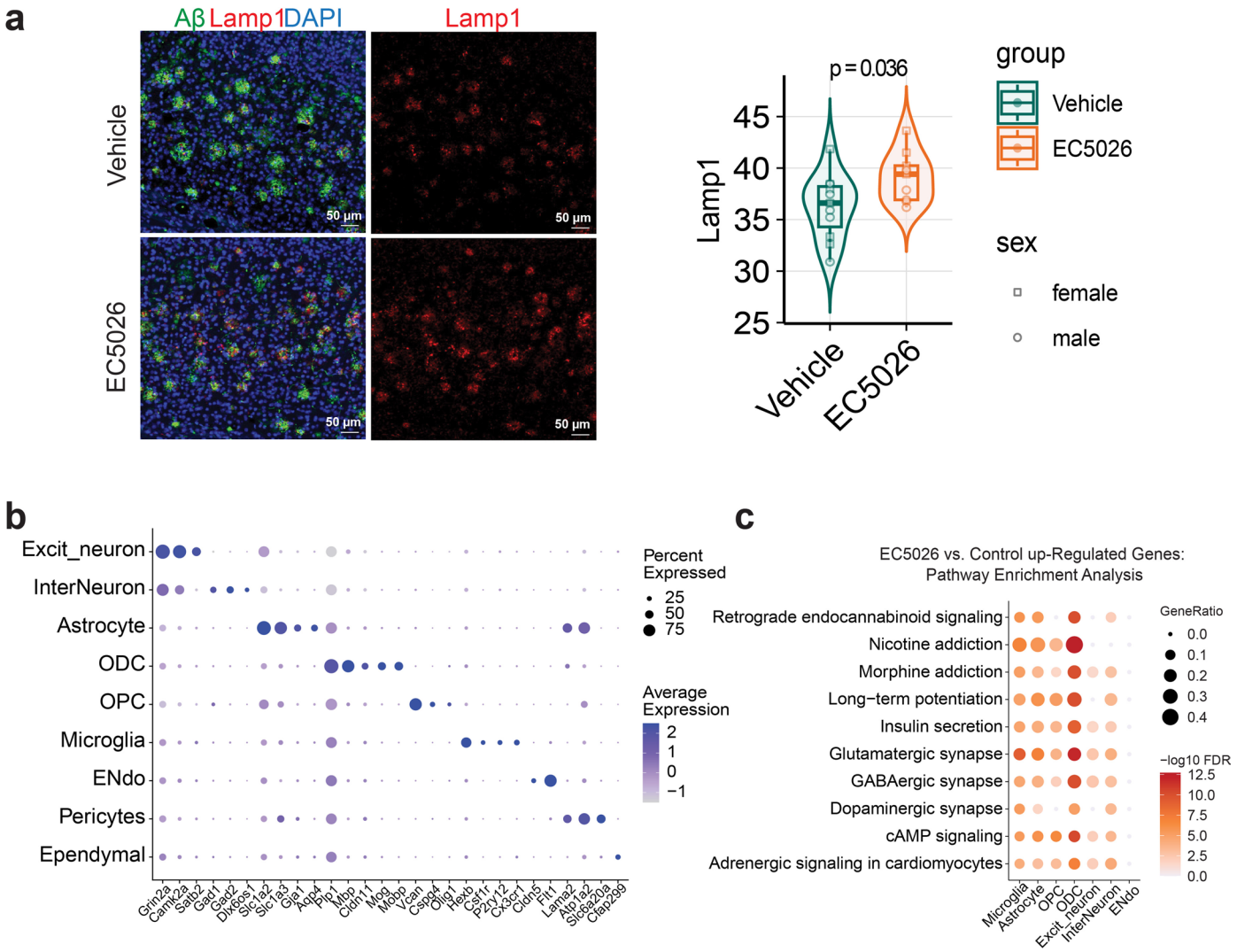
Extended Data Fig. 3 | Summary of results across five Mendelian randomization (MR) methods. a, The upset plot showing the intersections of significant drug targets among 5 MR models. **b**, The bar plot shows alzMR score for 19 drug targets across 5 MR models for EA. **c**, The bar plot shows alzMR score for 7 drug targets across 5 MR models for AA.

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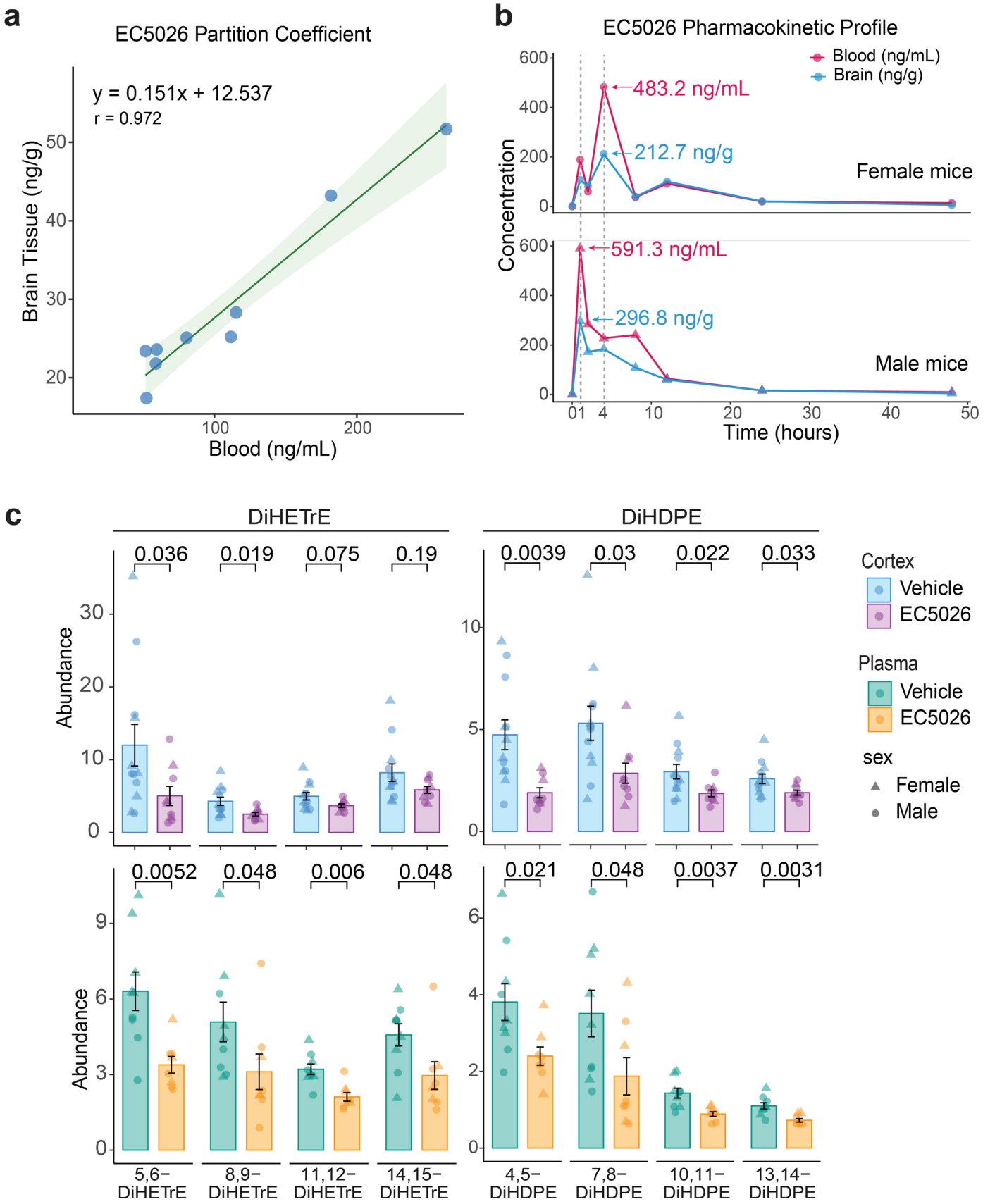
Extended Data Fig. 4 | Analysis of the relationships among the SNPs at the CLU locus. a, Conditional analysis of EPHX2 pQTL mutations. Original refers to the -$\log_{10}$(p-value) from the originally published GWAS summary statistics; conditional refers to the -$\log_{10}$(p-value) after conditional analysis based on SNP rs1532278. **b**, Multi-causal joint estimate of four SNPs at CLU locus. The black dashed line marks genome-wide significance ($P = 5 \times 10^{-8}$) and the grey dashed line marks $P = 1 \times 10^{-5}$. **c**, The scatter plot shows conditional association analysis at the CLU locus. **d**, CRISPRi of rs1532278 and rs2741342 does not affect PTK2B expression in iPSC-derived neurons (two independent gRNAs, 3 replicates each; n = 6 per group). Box plots show the median (center line), 25th-75th percentiles (box bounds) and whiskers extending to 1.5 × the interquartile range (IQR); dots are individual measurements; P values are from two-sided Student's t-tests.

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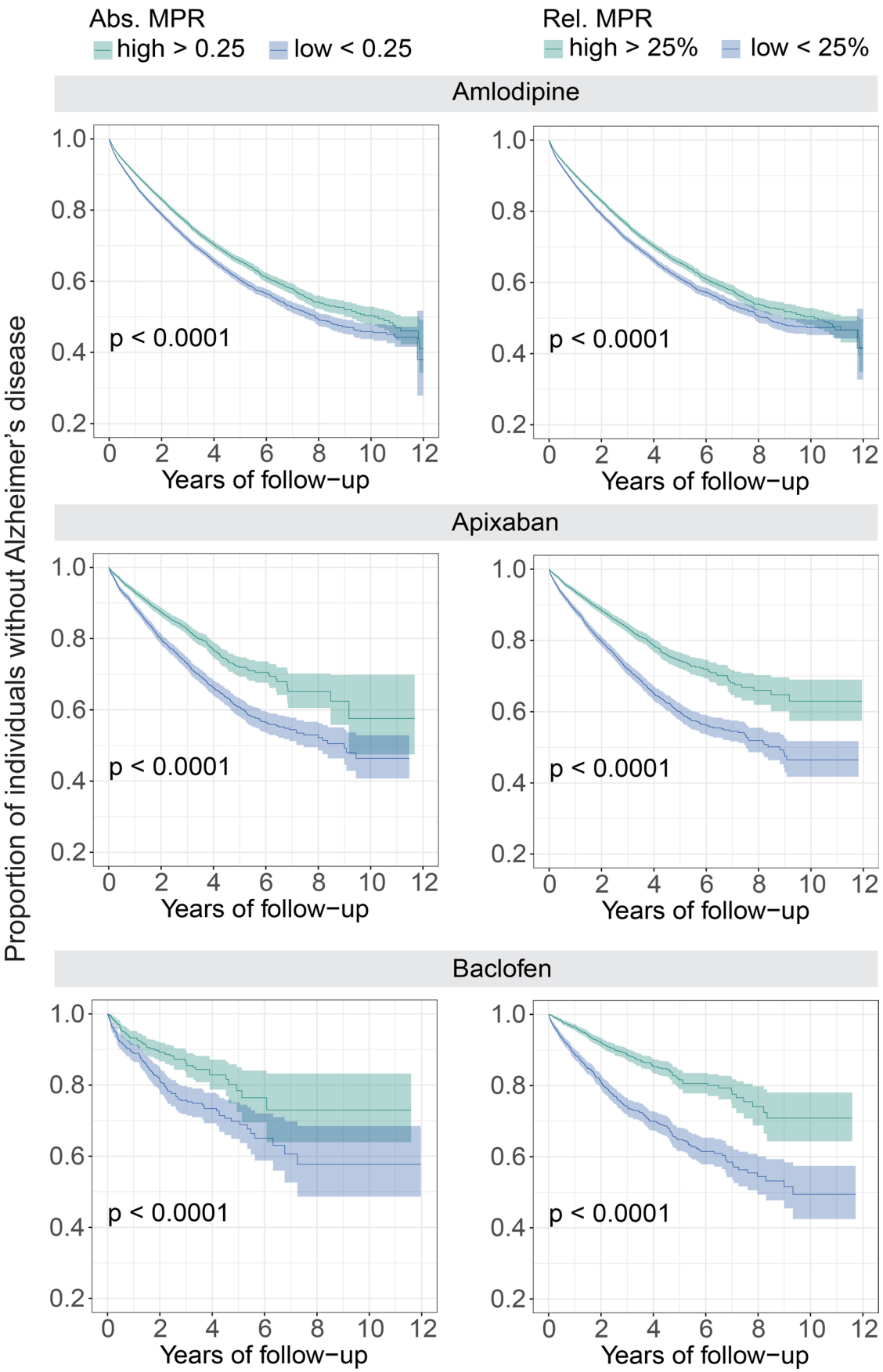
Extended Data Fig. 5 | Single-nucleus transcriptome effects of EC5026 in 5xFAD AD mouse model. a, The immunostaining results for microglia lysosomal function. **b**, Dot plot showing the marker genes for each cell type of mouse brain. **c**, Pathway enrichment analysis for up-regulated genes comparing EC5026-treated samples to controls across 7 cell types. Pathway enrichment used a one-sided hypergeometric test with Benjamini-Hochberg FDR correction.

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Extended Data Fig. 6 | Pharmacokinetic (PK) and pharmacodynamic (PD) analysis of EC5026 treatment in 5xFAD mouse model of AD. a, Brain-to-blood partition analysis of EC5026 showing the relationship between blood and brain concentrations. **b**, Pharmacokinetic profile of EC5026 showing brain and plasma concentrations over time in female and male mice. **c**, Pharmacodynamic analysis of EPHX2 downstream metabolites, showing levels of diols (DHETs) in brain tissue and plasma following EC5026 treatment. EC5026 n = 9, vehicle n = 12; data are mean ± SEM; P values from two-sided Student's t-tests.

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Extended Data Fig. 7 | Drug exposure and AD-free survival in patients with MCI. The curve displays the proportion of individuals who did not develop AD for amlodipine, baclofen and apixaban under two DWAS exposure models. MPR, medication possession ratio Abs: high exposure (MPR > 0.25) vs. low exposure (MPR ≤ 0.25). Rel: relative high exposure (MPR > 25% quantile of the corresponding drug's MPRs) vs. relative low exposure (MPR ≤ 25%). Shaded bands indicate the 95% confidence interval; P values are from a two-sided log-rank test.

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☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	We used the OPTUM Clinformatics Database (years 2007-2020, > 80 million individuals, and 5.6 million older individuals (≥60 years) with ≥3 years continuous enrollment) and the MarketScan Claims database (172 million insured individuals [Individuals were enrolled into either Medicare, Medicaid, or other Commercial health insurers from January 2011 to December 2020]), two commercially insured US individuals and the US Medicare Advantage beneficiaries. The H3K4me3 and H3K27ac Chip-seq data of Hippocampus can be accessed at GSM773022 and GSM773020; and the H3K4me3 and H3K27ac Chip-seq data of inferior temporal lobe can be accessed at GSM772996 and GSM772995. The single-cell ATAC-seq and HiChIP can be accessed at GSE147672. The bulk-RNAseq data is available from ROSMAP at syn3219045. For the details of the datasets used in this study are provided in Supplementary Tables 1 and 2. Noldus EthoVision XT 15: behavioral tracking for novel object recognition, Morris water maze. Video Freeze (Med Associates, Inc) for Contextual Fear conditioning. Image J 1.53K for western blot and immunohistochemistry data analysis All custom codes for data collections and analyses are free available: https://github.com/ChengF-Lab/alzMRTargets
Data analysis	Python 3.10.2; NumPy 1.26.4; pandas 2.3.3; R v4.1.1; TwoSampleMR v 0.5.7; MendelianRandomization; MR-PRESSO; PLINK v1.9; coloc v 5.2.3; FOCUS v 0.6.10; survival v 3.4.0; CellRanger v7.0.0; Seurat v4.0.2; ggplot2 4.0.2. Github package for newly developed method described/linked in Code Availability section of manuscript document.

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All custom codes for data collection and analyses, and datasets are free available: https://github.com/ChengF-Lab/alzMRTargets

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	We matched sex and gender using propensity-score matching approaches in all patient data analyses.
Population characteristics	We used R package comorbidity to define various patient characteristics, including cerebrovascular disease, diabetes, heart disease, liver disease, peripheral vascular disease, pulmonary disease, renal disease, and rheumatoid arthritis. We collected demographic variables (e.g., sex, year of birth, and race), administrative information (e.g. insurance enrollment record), diagnoses, procedures, and pharmacy claims, from the OPTUM Clinformatics Database (years 2007-2020) and the MarketScan Claims database (January 2011 to December 2020), two commercially insured US individuals and the US Medicare Advantage beneficiaries. We used European ancestry and African ancestry as the grouping variable. Ancestry was defined in the public available source GWAS and eQTL datasets. In the electronic health record cohorts, race (e.g., Black, White) was recorded as self-reported administrative data. We analyzed ancestry groups separately because AD prevalence, risk factors, drug response, and available genetic data differ between EA and AA populations. Ancestry/race was not used as a proxy for any other variable. In all drug-outcome analyses we adjusted for age, sex, race, and 21 AD-related comorbidities, and applied 1:1 propensity-score matching to control for confounding.
Recruitment	We didn't recruited new patients in this manuscript. We collected demographic variables (e.g., sex, year of birth, and race), administrative information (e.g. insurance enrollment record), diagnoses, procedures, and pharmacy claims, from the OPTUM Clinformatics Database (years 2007-2020, > 80 million individuals, and 5.6 million older individuals (≥60 years) with ≥3 years continuous enrollment) and the MarketScan Claims database (172 million insured individuals [Individuals were enrolled into either Medicare, Medicaid, or other Commercial health insurers from January 2011 to December 2020], two commercially insured US individuals and the US Medicare Advantage beneficiaries. Genetic analyses used publicly available GWAS and eQTL/pQTL datasets. The drug-repurposing analysis used 111,680 patients with mild cognitive impairment (MCI) and no prior AD, cancer, hemiplegia/paraplegia, or Parkinson's disease; covariates included age, sex, race, and 21 AD-related comorbidities.
Ethics oversight	We used the de-identified health insurance claims data. There is no any ethics issues.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We collected AD summary statistic datasets from seven case-control GWAS studies, including five AD GWAS datasets from EAs and two AD GWAS datasets from AAs (All details are provided in the Supplementary Table 2): 1) AD Bellenguez, 2022, (GWAS catalog ID, GCST90027158): this dataset includes 85,934 clinically diagnosed or proxy AD cases and 401,577controls. 2) LOAD.NoUKB23 Wightman, 2021 is a GWAS study for late onset AD (LOAD). In this study, data from UK Biobank and 23andMe (UKB23) were excluded from the GWAS summary statistic data by following human subject study polices from UK Biobank and 23andMe. The final cohort included 39,918 LOAD cases and 358,140 controls. 3) AD Schwartzentruber, 2021 (GWAS catalog ID, GCST90012877) includes 75,024 cases who diagnosed AD or have family history of AD, and 397,844 controls. 4) Family.Htx.AD Schwartzentruber26, 2021 (GWAS catalog ID, GCST90012878) includes 53,042 family history of AD cases and 355,900 controls. 5) AD Jansen, 2019 (GWAS catalog ID, GCST007320), this cohort includes 71,880 cases of LOAD and family history of AD, and 383,378 controls. 6) AD-AAs Kunkle 2020 were accessed from the National Institute on Aging Genetics of Alzheimer's Disease Data

Storage (NIAGADS), including of 2,784 AD cases and 5,222 cognitively normal controls.

In this study, we utilized three pQTL and nine eQTL (expression quantitative trait locus) datasets of human brain samples from multiple biobanks (All details are provided in the Supplementary Table 1). Three pQTL datasets of human cortex samples were generated from ROSMAP including 1) a pQTL datasets with AD and control subjects and 2) a pQTL dataset with control subjects only; and 3) a third pQTL dataset from Banner brain projects. The following nine cis-eQTL instruments sources are publicly available from the AD Knowledge Portal84 (Synapse ID: syn17015233) and Metabrian. Three eQTL datasets of cortex samples were generated by 4) ROSMAP (Religious Orders Study/ Memory and Aging Project), 5) Mayo clinic biobank and 6) Metabrain24. Two eQTL datasets from cerebrum samples were generated from 7) Mayo clinic and 8) Metabrain. 9) A meta-analysis eQTL dataset . 10) A eQTL dataset of BasalGanglia samples, 11) hippocampus samples and 12) spinal cord samples collected from Metabrain. The single-cell ATAC-seq and HiChIP can be accessed at GSE147672. The bulk-RNAseq data is available from ROSMAP at syn3219045.

The pharmacoepidemiology study used OPTUM Clinformatics Data (years 2007-2020) the MarketScan Claims database (172 million insured individuals [Individuals were enrolled into either Medicare, Medicaid, or other Commercial health insurers from January 2011 to December 2020]). The data were derived from commercially insured US individuals and the US Medicare Advantage beneficiaries. It provided demographic variables (e.g., sex, year of birth, and race), administrative information (e.g. insurance enrollment record), diagnoses, procedures, and pharmacy claims. The data included > 80 million individuals, and 5.6 million older individuals (≥60 years) with ≥3 years continuous enrollment. We used R packages Matchit and optmatch to derive 1:1 propensity score-matched (PS-matched) datasets for all de-identified patient data analysis. The propensity score model included age, sex, race and 21 comorbidities. We used the Kaplan-Meier estimator to estimate probability of not developing AD over time. We used log-rank test to compare the probabilities of not developing Alzheimer's disease (AD) in each PS-matched dataset.

Data exclusions: No data exclusions were applied.

Replication: We perform cross brain biobanks-validation observations.

Randomization: The mice were randomly assigned to EC5026 or vehicle treatment groups.

Blinding: For all animal experiments, different mouse group are blinded during behavior and other testing.

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We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

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☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Western blot (iPSC neurons): pTau-181 (Cell Signaling #12885, 1:1000); Tau (Cell Signaling #46687, 1:1000); FLAG (Sigma-Aldrich #F3165, 1:2000); GAPDH (Cell Signaling #97166, 1:2000).
Immunofluorescence (iPSC neurons and human brain organoids): anti-β-III-tubulin Alexa Fluor 488 conjugate (Abcam, 1:100); anti-Amyloid (B-4) Alexa Fluor 488 conjugate (Santa Cruz #sc-28365, 1:200); anti-Tau (phospho-T205) Alexa Fluor 647 conjugate (Abcam #AB307376, 1:200).
Immunofluorescence (mouse brain): rabbit anti-Iba1 (Fujifilm Wako #019-19741, 1:500); mouse anti-amyloid (BioLegend #803001, 1:500). Secondaries: donkey anti-mouse Alexa Fluor 488 (Thermo Fisher #A-21202, 1:1000); donkey anti-rabbit Alexa Fluor 568 (Thermo Fisher #A10042, 1:1000).

Validation: All antibodies were validated based on original company protocoles before used.

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s): Forebrain neurons were differentiated from a LOAD patient iPSC (female [63 years old], # IUGB269.1) provided by the National Centralized Repository for Alzheimer's Disease and Related Dementias (NCRAD) at the Indiana University. We differentiated the iPSC into mature forebrain neurons using a commercial pipeline from STEMCELL Technologies. iPSCs were initially cultured in mTeSR™1 medium, dissociated, and plated on Matrigel-coated dishes. They were cultured in neural

induction medium (Catalog # 05839) to form neural progenitor cells (NPCs). The NPCs were later cultured in forebrain neuron differentiation (Catalog # 08600) and maturation media (Catalog # 08605). After 10 days of maturation, the neurons were treated with 2 µg/ml doxycycline for 48 hours. This process was conducted in a controlled environment of 37°C and 5% CO2.

i3N-WTC11 induced neurons were generated using a two-stage differentiation procedure, as described previously. To achieve stable and uniform expression of transgenes during neuronal maturation, a dCas9-KRAB construct under the control of the CAG promoter was inserted into the intronic region of the CLYBL safe harbor locus. Single guide RNAs (sgRNAs) directed against prioritized SNP or risk gene promoters were designed using CHOP-CHOP. To ensure sgRNA specificity and minimize off-target effects, we evaluated potential off-target binding sites containing one base mismatch, two base mismatches, or three base mismatches relative to the intended target sequence. For cloning, sgRNA oligonucleotides were inserted into the lentiGuide-puro backbone (Addgene, Plasmid #52963) and confirmed with Sanger sequencing. Lentiviral particles were produced in 293T cells by co-transfection of the sgRNA plasmid with pMD2.G (Addgene, Plasmid #12259) and psPAX (Addgene, Plasmid #12260) helper vectors using Lipofectamin 2000 (Invitrogen Catalog No. 1166802). Viral supernatants were harvested over 48 h, passed through 0.45 µm filters (Millipore Catalog #SLHV033RS), and concentrated with Amicon Ultra centrifugal devices (Millipore Catalog #UFC801024). Viral titers were determined in iPSCs by puromycin selection. For functional CRISPRi experiments, iPSCs were differentiated into excitatory neurons for 14 days before infection with lentiviruses carrying sgRNAs at a multiplicity of infection (MOI) of approximately 5. Following transduction, cultures were maintained under puromycin selection (0.5 µg/mL) for four days. Cells were harvested seven days after infection for RNA extraction. Gene expression changes were quantified by RT-qPCR from at least 3 biological replicates, and relative expression across replicates was analyzed by two-tailed Student's t-test.

Authentication	The iPSC lines were authenticated by the National Centralized Repository for Alzheimer's Disease and Related Dementias (NCRAD) at the Indiana University.
Mycoplasma contamination	The iPSC line and derived neurons were checked by Cell Core at the Lerner Research Institute at the Cleveland Clinic and there is no any Mycoplasma contamination issue before used.
Commonly misidentified lines (See ICLAC register)	No.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Species: mouse. Strain: 5xFAD transgenic (RRID:MMRRC_034840-JAX / Jax 006554; carrying APP K670N/M671L, I716V, V717I and PSEN1 M146L, L286V), from Jackson Laboratory, purchased at 6-8 weeks of age. Treatment started at 8 months of age for 3 months (EC5026 3 mg/kg/day in drinking water, or 1% PEG-400 vehicle). Mice were group-housed at 20-22 °C, 30-70% humidity, on a 12h light/12h dark cycle, and tested in the light phase.
Wild animals	not applicable
Reporting on sex	Both female and male 5xFAD mice were used.
Field-collected samples	Mice were purchased from Jackson Laboratories (Bar Harbor, ME) at 6-8 weeks of age. All mice were housed in a designated animal facility at 20-22°C on a 12-h light/dark cycle with food and water available ad libitum. All protocols were approved by the Cleveland Clinic Lerner Research Institute Institutional Animal Care and Use Committee (IACUC) in compliance with the guidelines established by the Public Health Service Guide for the Care and Use of Laboratory Animals.
Ethics oversight	All mouse studies protocols were approved by the Cleveland Clinic Lerner Research Institute Institutional Animal Care and Use Committee (IACUC) in compliance with the guidelines established by the Public Health Service Guide for the Care and Use of Laboratory Animals.

Note that full information on the approval of the study protocol must also be provided in the manuscript.